

Restoration is a partial but not complete solution for long-term declines in oyster populations—A case history from the northeastern Gulf of America

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Abstract

Objective: We sought to evaluate whether oyster reef restoration is associated with sustained increases in intertidal eastern oyster *Crassostrea virginica* density in Suwannee Sound, Florida, and to examine how commercial fishing activity and freshwater discharge may modify those patterns.

Methods: We monitored intertidal oyster reefs for 14 years, including 9 years prior to large-scale restoration (completed in 2018) and 5 years of standardized postrestoration monitoring beginning in 2019 at Lone Cabbage Reef, an offshore intertidal reef complex that was rebuilt to target elevation using dolomite limestone. A Bayesian negative-binomial generalized linear mixed model framework estimated oyster densities while accounting for spatial differences among inshore, nearshore, and offshore reefs; variation in transect length; and temporal variability. We assessed restoration effectiveness via repeated-measures monitoring of fixed restored sites and landscape-scale comparisons of restored versus unrestored reefs. We evaluated covariates of management interest—commercial fishing activity and river discharge—using model comparison and posterior predictions of oyster density at observed covariate values.

Results: Restoration produced large per-area gains in live-oyster density on offshore reefs. At four fixed restored offshore stations, mean densities rose from near 0 to 91 oysters/m² (95% credible interval [CrI] = 43–185) within 6 months and reached 244 oysters/m² (95% CrI = 134–433) by year 5 at fixed restoration stations. At the broader landscape scale, restored offshore reefs averaged about 83 oysters/m² (95% CrI = 26–157) across sites and postrestoration periods, whereas model-based estimates for restored offshore reefs reached 218 oysters/m² (95% CrI = 128–366) in the final sampling period.

Conclusions: Restoring reef substrate rebuilt the physical capacity for oysters to persist and grow at Lone Cabbage Reef, yielding sustained density gains that degraded sites did not achieve on their own. Harvest status and river discharge showed weak, inconsistent associations once restoration was included, consistent with the interpretation that oyster cultch provides structural capacity, while environmental and fishing conditions modulate around it. Rebuilding and conserving oyster cultch constitute the essential first step to repair and sustain degraded oyster reefs; fishing regulations or favorable environmental conditions alone are unlikely to replace the need for reef structure.

Keywords: adaptive management, cultch limitation, eastern oyster, oyster restoration, reef elevation

Lay summary

Large-scale oyster reef restoration in Florida produced sustained density gains that degraded reefs could not achieve alone. Cultch addition at an appropriate elevation was the dominant driver; river discharge and commercial fishing had comparatively minor effects.

Introduction

Signs are increasing that many large eastern oyster *Crassostrea virginica* populations in the Gulf of America (also known as “Gulf of Mexico”; hereafter, “Gulf Coast”) have collapsed. Indicators include declining fishery landings, trends in fishery-independent surveys, and allocation of more than US\$200 million for restoration (Deepwater Horizon Natural Resource Damage Assessment Trustees, 2017; Du et al., 2021; Moore et al., 2020; Pine et al., 2023). This collapse is alarming given the ecological, economic, and cultural importance of this reef-building species. Oysters and their shell substrate provide habitat for fish, invertebrates, and shorebirds, along with shoreline protection, nutrient cycling, and salinity buffering (Coen et al., 2007; Grabowski et al., 2012; Kopecky et al., 2023; Lenihan, 1999; Ridge et al., 2017). Oysters are also culturally important seafood, having been harvested for thousands of years on the Gulf Coast (Barbour et al., 2019; Eleuterius & Otvos, 1979; Holmes & Trickey, 1974).

The commercial and ecosystem value of oysters is widely recognized, and oyster restoration is expanding across the Gulf Coast (Bendick et al., 2018; La Peyre et al., 2014; Pine et al., 2022; Smith et al., 2023). Outcomes of these efforts are mixed, and many oyster restoration projects regionally remain unevaluated or unsuccessful (Bersoza Hernández et al., 2018; Blomberg et al., 2018; La Peyre et al., 2022; Pine et al., 2023). In some systems, oyster populations appear to be near or past ecological tipping points, where recovery is difficult even after stressors are reduced (Beisner et al., 2003; Camp et al., 2026; Johnson et al., 2022, 2023; Pine et al., 2015). For example, Pine et al. (2023) reported failure of oyster populations to recover across six restoration projects in three of Florida’s Gulf coast estuaries despite restoration and fishing restrictions, and Johnson et al. (2022) and Camp et al. (2026) showed through simulation how hysteresis can arise from interactions among oysters, substrate, and fisheries, similar to dynamics observed in collapsed north Atlantic Cod *Gadus morhua* stocks (Hutchings, 1995; Walters & Maguire, 1996). These findings underscore the need to identify the drivers of restoration outcomes and ways to avoid collapse.

Restoration outcomes are strongly shaped by broader environmental and management stressors. Across the Gulf Coast, oyster declines have been linked to overharvest, inadequate shell replacement, altered freshwater flows, drought, disease, and episodic disturbances, such as the Deepwater Horizon oil spill (Du et al., 2021; Gledhill et al., 2020; Petes et al., 2012). These factors operate across multiple spatial and temporal scales and are further influenced by climate and land use change, thus complicating attribution and the extent to which any single driver is manageable. Learning from unsuccessful restoration projects has also been constrained by weaknesses in project design, inconsistent or short-term postrestoration monitoring, and uncertainty about whether poor outcomes reflect reef design, environmental conditions, or both

(Blomberg et al., 2018; Pine et al., 2023). In Florida, conflicts over freshwater management and oyster harvest have led to more than \$100 million in litigation among states over freshwater inputs to Apalachicola Bay (Barnett, 2021; Kelly, 2019). Together, these examples highlight the difficulty of disentangling stressors to inform water policy and fishery regulation, and they emphasize the need to test whether restoration can succeed even amid broader anthropogenic change.

Informing oyster restoration and fishery management in the Gulf of America is both urgent and complex. In some estuaries, oyster populations decline gradually; in others, they collapse for unclear reasons (Camp et al., 2015; Moore et al., 2020; Pine et al., 2015). Restoration has often been unsuccessful at reversing these trends, raising questions about the approaches required to restore viable reefs that provide ecosystem services and sustainable fisheries (Coen & Luckenbach, 2000; Grizzle et al., 2021; Hintenlang et al., 2023; Howie & Bishop, 2021; La Peyre et al., 2022; Lee, 2023).

Oyster restoration typically adds substrate (shell, limestone, or other durable materials) to promote larval settlement and rebuild three-dimensional reef structure (Baggett et al., 2015; Schulte et al., 2009). Substrate addition targets a common bottleneck in degraded systems: the loss of stable cultch that is needed for recruitment and reef persistence. However, where larval supply is low or environmental stressors suppress recruitment, substrate addition alone may not yield recovery. Disentangling substrate limitation from recruitment limitation is therefore essential for predicting when and where restoration will succeed (Powers et al., 2009; zu Ermgassen et al., 2016).

In the transitional area between the Florida panhandle and peninsula (regionally referred to as “Florida’s Big Bend”; Figure 1), partners from academia, government, and industry have collaborated for more than 14 years to monitor oysters and test restoration strategies. Initial assessments documented a 66% net loss of intertidal reef area between 1982 and 2010, with losses continuing thereafter (Moore et al., 2020; Seavey et al., 2011). Long-term monitoring showed that offshore reefs in Suwannee Sound remained in a degraded state for at least a decade, with persistently low densities and limited natural recovery (Moore et al., 2020).

These findings motivated active restoration at Lone Cabbage Reef, an offshore intertidal reef chain in Suwannee Sound (Figure 1). Pilot efforts demonstrated that oyster population recovery was possible with suitable substrate addition to recover lost cultch material, elevation, and structural complexity (Aufmuth et al., 2025), indicating substrate limitation rather than larval limitation on oyster recruitment (Frederick et al., 2016). Recent analyses in Suwannee Sound confirmed a strong nonlinear relationship between cultch mass and live-oyster density, with evidence of functional thresholds near 5–20 kg/m³, emphasizing that sufficient substrate is essential for oyster recruitment and viability (Casteel et al., 2025).

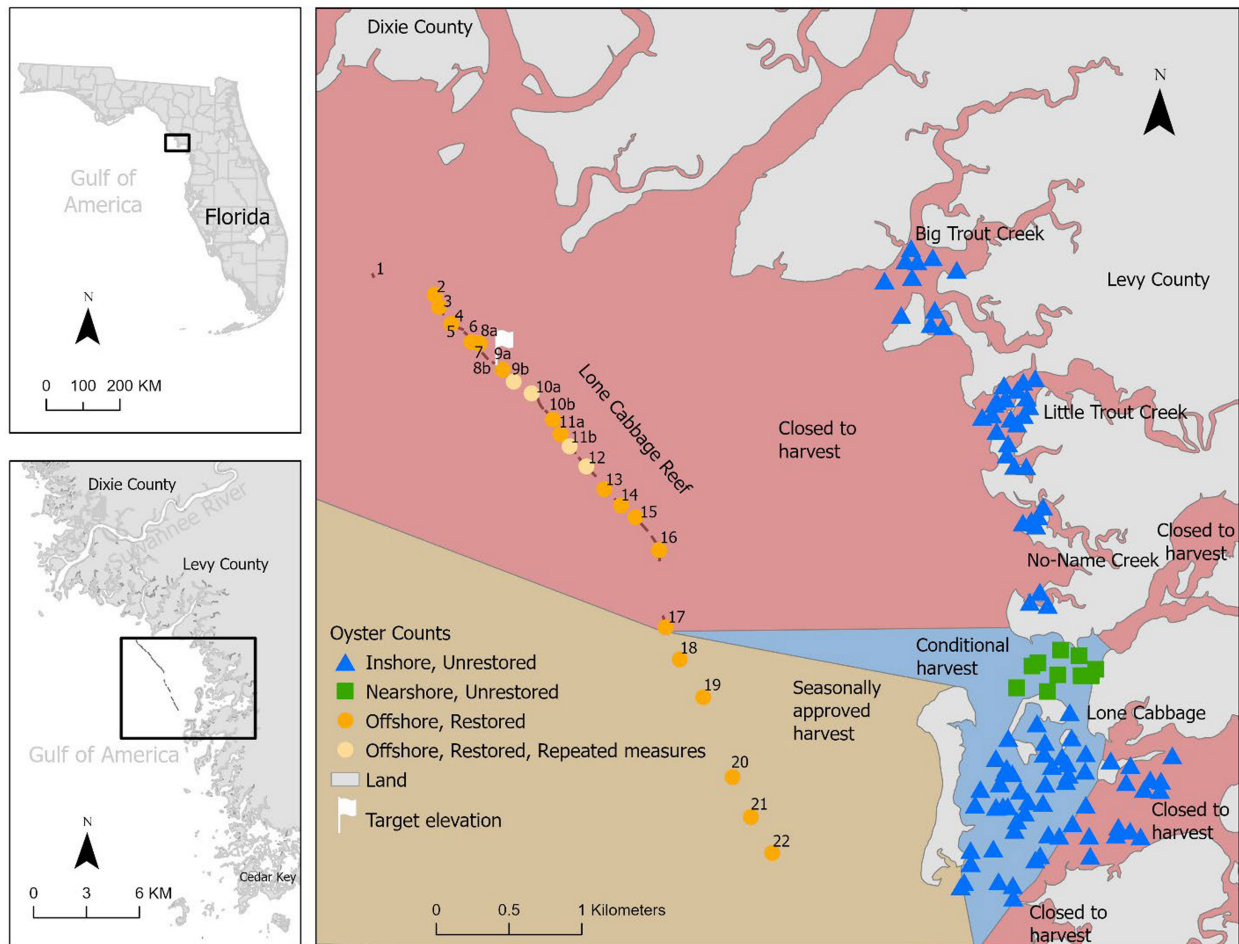


Figure 1 Study area for the Lone Cabbage Reef restoration project.

Building on this evidence, the large-scale restoration at Lone Cabbage Reef was implemented in 2019, covering a 5-km ribbon of relict reef with dolomite limestone and shell at target elevations matching nearby robust reefs (Aufmuth et al., 2025; Pine et al., 2022).

Spatial setting is a critical factor shaping oyster populations in the Big Bend region. In Suwannee Sound, intertidal reefs occur in inshore, nearshore, and offshore zones that trace relict shorelines from earlier sea levels (Hine et al., 1988; Raabe & Stumpf, 2016; Wright et al., 2005). These reef chains run parallel to the coast: Inshore reefs lie closest to the current shoreline, nearshore reefs form the next chain seaward, and offshore reefs are farthest from the current shoreline. Oyster densities vary along this gradient, with the highest densities occurring inshore and the lowest occurring offshore (Moore et al., 2020), likely reflecting geomorphic processes, wave energy exposure, sea level, and sediment mobility (Hine et al., 1988; Raabe & Stumpf, 2016).

We therefore included these strata explicitly in our analyses. Because restoration at Lone Cabbage Reef occurred exclusively at offshore sites, the zone with the most severe losses (Moore et al., 2020; Seavey et al., 2011), our evaluation of restoration effects explicitly accounted for spatial setting.

We focused on two management-relevant stressors in Suwannee Sound that may influence restoration outcomes and long-term oyster trends: river discharge and commercial

fishing activity. River discharge affects salinity and nutrient conditions in ways that can cut in both directions: High discharge depresses salinity below oyster tolerance thresholds, suppressing recruitment and survival, while low discharge allows salinity to rise to levels that favor predators and parasites, and both extremes can reduce oyster density (Du et al., 2021; Fisch & Pine, 2016; Moore et al., 2020). Commercial fishing directly removes live oysters and shell substrate from subtidal reefs, reducing cultch availability and potentially disrupting source-sink dynamics through which subtidal reefs may supply recruits to adjacent intertidal populations (Moore et al., 2020). Both factors are regionally important, measurable, and actively debated in management decisions, even for intertidal reefs where oysters rarely reach legal harvest size (Pine et al., 2026). Inclusion of river discharge and commercial fishing activity in our models allowed us to test whether these stressors altered oyster population trends beyond the direct effect of restoration and to compare their relative influence in terms of expected oysters per square meter.

Here, we used 14 years of oyster monitoring that spanned pre- and postrestoration periods to evaluate population responses to large-scale restoration at Lone Cabbage Reef. We addressed four questions: (1) how live-oyster counts changed across reef zones before pilot and full restoration; (2) how counts changed through time at fixed restored offshore

transects; (3) how average oyster densities compared among inshore, nearshore, and offshore reefs and between restored and unrestored sites; and (4) whether river discharge and commercial fishing activity explained variation in oyster counts beyond the effect of restoration.

By documenting a large-scale restoration conducted with local oyster harvesters, this study provides a counterpoint to recent work showing resistance to restoration in Florida (Hintenlang et al., 2023; Pine et al., 2023). Because restoration was implemented at an ecologically appropriate elevation with locally sourced material, this case also helps to evaluate whether restoration design contributes to differences among outcomes across Florida systems. More broadly, the results clarify when substrate restoration may overcome long-term declines and contribute to recovery of an economically, ecologically, and culturally important resource.

Methods

Study site

Suwannee Sound is a shallow, drowned-lagoon, open-ocean-facing estuary in the northeastern Gulf of America (Figure 1). The region is often considered to be of high quality due to its low human population density and its extensive coastal habitat, which is protected as state or federal conservation lands (Mattson, 2002; Raabe & Stumpf, 2016). Major economic activities include commercial harvest from wild-oyster fisheries, hard clam *Mercenaria mercenaria* and oyster aquaculture, timber harvest, recreational fishing, and water-based recreation (Southwick Associates, 2021).

Suwannee Sound has low wave energy and minor elevation gradients, and oyster reefs have been a persistent landscape feature for thousands of years (Hine et al., 1988; Raabe et al., 2004). Many reefs are intertidal and dewater at different points in the tide cycle depending on reef and tidal elevation, whereas subtidal reefs occur in gaps between intertidal reefs and remain submerged. Our monitoring and restoration work in this region began in 2010 and focused on intertidal reefs (Frederick et al., 2016; Moore et al., 2020; Seavey et al., 2011), particularly Lone Cabbage Reef, a reef chain that is oriented parallel to the coastline (Aufmuth et al., 2025). Lone Cabbage Reef is at least 2,000 years old (Grinnell, 1972; Wright et al., 2005); it supported robust oyster populations at least through the 1980s and has subsequently declined (Seavey et al., 2011). Historically, the Suwannee Sound region supported Florida's second-largest oyster fishery based on trips and landings, and it became the state's largest oyster fishery after the 2012 collapse of the Apalachicola Bay fishery (Florida Fish and Wildlife Conservation Commission, 2025).

Restoration

Restoration of Lone Cabbage Reef occurred in two phases. Phase 1 was a pilot project that placed 360 m³ of dolomite limestone at multiple sites, established proof of concept for this restoration approach, and indicated that the site was not limited by the supply of oyster larvae (Frederick et al., 2016). Four site pairs, each including one restored reef and one similar-sized control reef, were surveyed before restoration in

2013 and after restoration in 2015; the oyster population response was significant (Frederick et al., 2016). Phase 2, completed in 2018, restored 26 reefs with approximately 13,000 m³ of dolomite limestone; standardized postrestoration monitoring began in 2019 (Aufmuth et al., 2025).

Restoration methods were similar in both phases. High-resolution, on-water surveys delineated the elevation and footprint of relict reefs. Real-time kinematic positioning surveys of adjacent, apparently robust wild reefs established target restoration elevations, and volumetric calculations determined the quantity of rock that was required to build each reef to that target (Aufmuth et al., 2025). The initial target elevation was −0.442 m NAVD88 (North American Vertical Datum of 1988), based on viable reefs at the northern end of Lone Cabbage Reef ("target elevation" location in Figure 1). To allow natural variation and to increase reef complexity, final target elevations ranged from −0.5944 to −0.3658 m NAVD88 (Aufmuth et al., 2025). Pilot project reefs were incorporated into the larger effort by adding rock so that all restored sites received a similar treatment and were brought to comparable elevations (Aufmuth et al., 2025).

The restored reef complex spanned areas with different harvest regulations. Both before and after restoration, the northern two-thirds of the reef were within areas closed to commercial harvest year-round, whereas the southern one-third occurred in areas that were seasonally open to harvest. Wild-oyster harvest was permitted from September 1 through May 31 (closed from June 1 to August 30) and was subject to a minimum legal shell height of 76.2 mm. Harvest was also regulated spatially through designated open and closed zones that included inshore, nearshore, and offshore reef areas (Figure 1). Additional harvest restrictions based on routine water quality testing also occurred, including closures during periods of high river discharge. Inshore and nearshore reefs in the study area were similarly subject to these spatial and seasonal regulations, with harvest status varying by zone and year.

Data collection

We used low-tide line-transect surveys on exposed reefs to track spatial and temporal patterns in live-oyster counts in Suwannee Sound before and after restoration of Lone Cabbage Reef. Methods followed prior monitoring and restoration work in this region (Frederick et al., 2016; Moore & Pine, 2021; Moore et al., 2020; Seavey et al., 2011). Sites were selected using a combination of fixed-location and random sampling approaches (described below). The oyster-counting protocol was identical across all sites, although transect placement differed slightly between restored and unrestored reefs to capture comparable reef profiles (see *Random-location site selection*).

At each location, transect start and end points were located using a Juniper Systems Geode GNSS receiver (Juniper Systems, Logan, Utah) that was wirelessly connected to a handheld field computer for real-time data acquisition. End points were temporarily marked with metal H-shaped frames hammered into the reef to define a belt transect; frame width (15.24 cm) set the transect width. Strings threaded between

frames on both sides provided visual guides for counting from above. Oysters were counted in situ without disturbance or collection, and a metric tape between endpoints provided repeatable reference points along the transect.

All oysters with more than 50% of the shell within the transect were counted. Trained observers ([Appendix A](#)) recorded live and dead oysters in 2.5-m segments. To minimize misidentification and allow for nonlethal field counts, we set a minimum shell height of approximately 18 mm, using a U.S. dime as a visual reference. Only oysters visible on the surface were counted; we did not excavate for buried oysters. Oysters within clumps were counted without damaging the clump. Segmented recording reduced counting fatigue and errors from losing position on high-density reefs and allowed multiple observers to complete different segments within a single transect, improving efficiency during tide-limited sampling windows. Approximately 20% of transects were double-counted, usually by a different observer.

Fixed-location site selection

Four individual reefs in the Lone Cabbage chain were sampled repeatedly before and after restoration. Three replicate transects were sampled at each of the four stations across nine occasions: summer 2013, winter 2014–2015, summer 2015, winter 2017–2018, winter 2018–2019, winter 2019–2020, winter 2020–2021, winter 2021–2022, and winter 2022–2023 ([Tables B1](#) and [B2](#)). Transect locations and methods were repeated to instrument precision on each visit.

Random-location site selection

To assess whether spatial and temporal patterns were related to restoration, we randomly sampled both restored and unrestored reefs across Suwannee Sound. Using existing GIS layers of oyster reef distribution ([Radabaugh et al., 2019](#)), we identified potential line-transect locations and then randomly selected transects from the full set of possibilities. Sites were classified by two crossed factors: restoration status (restored or unrestored) and harvest status (open or closed to commercial harvest). Not all combinations of restoration and harvest status were represented in every period. Inshore unrestored reefs were sampled in both open and closed harvest areas throughout the study. Nearshore unrestored reefs were sampled only in areas open to harvest. Offshore unrestored reefs were sampled only in areas closed to harvest during periods 1–16, with one exception in period 19. Offshore restored reefs were sampled in both open and closed harvest areas from period 18 onward. Before each field season, we used a power analysis to determine the total meters of transect to sample within each combination of these factors ([Moore & Pine, 2021](#)), using counts from the previous season or all available counts for that combination—whichever produced the more conservative estimate. Midseason, we adaptively updated sampling effort by rerunning the power analysis with observed counts to allocate more effort to strata with higher variance ([Moore & Pine, 2021](#)).

Transect placement differed slightly between reef types to capture comparable elevation profiles. On restored reefs, multiple transects were established across the narrowest part

of each reef, generally oriented east–west across the reef profile. On unrestored reefs, transects were placed to cross the highest point of the reef ridge so as to capture the full elevation gradient. In both cases, the oyster-counting protocol was identical; only the placement approach differed to account for differences in reef geometry.

Distinguishing restoration effects from site and temporal variation

We estimated restoration effects while accounting for site location and sampling period using two complementary sampling designs. First, stratified random sampling provided broad spatial coverage and supported comparisons among reef zones (inshore, nearshore, or offshore) and restoration status (restored or unrestored). Because all restored reefs were offshore and no unrestored offshore reefs were sampled after large-scale restoration began, comparisons used (1) restored offshore reefs versus unrestored reefs in other zones and (2) restored offshore reefs versus offshore reefs sampled before restoration. Second, repeated sampling of fixed transects at four restored offshore sites provided a longitudinal design for evaluating within-site temporal change while holding site and transect location constant. Together, these designs allowed us to evaluate restoration responses while accounting for spatial and temporal heterogeneity in the study system.

Tidal height and season of sampling

All sampling occurred when tidal heights were below -0.24 m mean lower low water for at least 120 min, as predicted from National Oceanic and Atmospheric Administration station 8727520 at Cedar Key, Florida ([National Oceanic and Atmospheric Administration, 2025](#)). At these tide levels, large areas of intertidal reef are dewatered, permitting line-transect surveys on exposed reef surface. Suitable daylight conditions occurred on approximately 60 d/year, so we focused sampling on late October through March. We used R ([R Development Core Team, 2019](#)) with the packages *rtide* ([Thorley et al., 2021](#)) and *suncalc* ([Thieumel & Elmarhraoui, 2022](#)) to identify dates when predicted tides and sunrise permitted fieldwork; actual sampling windows sometimes differed from the predictions due to wind.

We organized time into “periods,” an integer factor used to simplify analysis ([Table B1](#)). Winters (October–March) were coded as even-numbered periods, and summers (April–September) were coded as odd-numbered periods. We included Suwannee Sound monitoring data from summer 2010 (period 1) through winter 2022–2023 (period 26; [Tables B1](#) and [B2](#)). Winter sampling reduced the variation associated with new larval recruitment and episodic mortality. Postrestoration sampling (i.e., following restoration in period 17) focused on restored reefs and unrestored reference reefs, accounting for harvest status (open or closed). We did not continue sampling unrestored offshore reefs after restoration, as earlier work indicated that these reefs were unlikely to recover without intervention ([Moore et al., 2020](#)); unrestored inshore and nearshore reefs served as reference sites in the postrestoration period.

Portions of these data have appeared in prior analyses (Frederick et al., 2016; Moore et al., 2020; Seavey et al., 2011). This is the first analysis to assess oyster population response to the full Lone Cabbage Reef restoration, evaluate the response through period 26, and place it in the context of trends on unrestored reefs across periods 1–26 (2010–2023).

Data analysis

We analyzed live-oyster counts using negative-binomial generalized linear mixed models with a log link. The response variable was the number of live oysters counted in each transect, with transect length included as a log offset so estimates are expressed on a per-meter basis. Temporal variation was modeled with a random intercept for period. In models where temporal deviations could differ among groups, the period effect was indexed by the observed period $\times$ treatment combination.

Fixed effects used a cell-means parameterization (no intercept) for each observed combination of reef zone (inshore, nearshore, or offshore) and restoration status (restored or unrestored), estimating a mean for each group without imposing a reference level. Posterior summaries are reported as means with 95% credible intervals (CrIs) on the response scale. Models were fitted in R using brms (CmdStan backend; Bürkner, 2017, 2021; Gabry et al., 2025) with weakly informative default priors, four chains of 3,000 iterations each, and 1,500 warmup iterations. Convergence required a convergence diagnostic (R-hat) no greater than 1.01 and large effective sample sizes, and all models met these criteria.

Independent variables included restoration status (restored or unrestored), reef zone (inshore, nearshore, or offshore), and harvest status (open or closed to commercial harvest). For covariate models (research question 4), continuous covariates were centered and scaled, and contrasts and figures are reported at the covariate mean.

For covariate analyses, we included river discharge (U.S. Geological Survey gauge station 02323500, Suwannee River at Wilcox, Florida; U.S. Geological Survey, 2025) and commercial fishery statistics (trips and landings for Levy, Dixie, and Taylor counties combined; C. Bradshaw, Florida Fish and Wildlife Conservation Commission, personal communication) as covariates, with interactions by reef type $\times$ restoration status. River discharge served as a proxy for salinity and nutrient delivery. To capture delayed effects, we evaluated mean daily discharge (AD) and total discharge (TD) for the current period and 1- and 2-year lags (Lag1, Lag2); the same lag structure was applied to trips and landings (Fisch & Pine, 2016; Pine et al., 2023). A site-only model served as the baseline. Candidate models were compared using Pareto smoothed importance sampling leave-one-out cross validation (PSIS-LOO; Vehtari et al., 2023), and effects are reported as differences in expected oyster density (oysters/m²) for contrasts of management interest.

Analyses were structured around four complementary questions addressing temporal trends, repeated-measures responses at restored sites, spatial contrasts among reef types, and the influence of environmental and fishery covariates. Each research question used the shared modeling framework described above, with minor modifications.

Research question 1: Landscape-scale temporal trends

How did live-oyster counts change across the landscape before restoration (periods 1–7), after the pilot restoration (periods 10–16), and after the full restoration (periods 18–26)?

To evaluate large-scale temporal trends in intertidal oyster populations, we specified a no-intercept model in which each reef type $\times$ restoration status combination was estimated independently. Because inshore and nearshore reefs were never restored, the model only included combinations observed in the data. We used posterior predictions to summarize oyster density trajectories across three phases: prerecovery (periods 1–7), post-pilot restoration (periods 10–16), and post-main restoration (periods 18–26). No data were collected during periods 8, 9, and 17 due to funding gaps or active construction. The full model structure is described in Appendix B.

Research question 2: Repeated-measures response at fixed restored sites

At Lone Cabbage Reef restoration sites that were sampled annually postrestoration, how did live-oyster counts change?

Four offshore sites that were restored during the pilot or full project were surveyed repeatedly following the summer 2018 restoration. Fixed transects were sampled beginning in winter 2018–2019 (period 18) and in each subsequent even-numbered period through 2023 (periods 20, 22, 24, and 26). Repeated sampling of the same transects minimized spatial variability, providing a fine-scale view of temporal trends, although results may not generalize to the full reef or other offshore sites. The model used a global intercept for mean density across restored offshore sites; posterior predictions were period-specific densities (Appendix B).

Research question 3: Spatial comparison of restored and unrestored reefs

Across Suwannee Sound, how did average live-oyster counts compare between restored and unrestored reefs while accounting for temporal and spatial variation?

This analysis used the same model structure as research question 1 (Appendix B; Figures B4 and B5) to estimate long-run mean oyster densities for each zone–status combination: inshore unrestored, nearshore unrestored, offshore unrestored, and offshore restored. The period random effect integrated observed temporal variation across sampling occasions, yielding conservative density estimates with uncertainty intervals that reflected both ecological variation and sampling gaps.

Research question 4: Influence of environmental and fishery covariates

We examined whether covariates of management interest—Suwannee River discharge and commercial fishing activity—explained variation in oyster counts between restored and unrestored reefs.

Although oysters on intertidal reefs rarely reach the legal harvest size, source–sink dynamics may link intertidal and subtidal populations. More broadly, the relationships among discharge, fishing activity, restoration, and oyster population response in Florida are complex (Fisch & Pine, 2016; Kelly, 2019; Moore et al., 2020; Pine et al., 2026). We also asked whether these covariates explained more or less variation than restoration itself.

We fitted two models. The main covariate model extended the landscape model from research questions 1 and 3 by adding covariates and their interactions with reef type, allowing covariate effects (slopes) to vary by reef zone and restoration status. Offshore unrestored observations were available only for periods before large-scale restoration (periods 1–3, 6–7, 10–11, and 16), so restoration contrasts used these prerestration observations as the unrestored offshore baseline.

A simplified restoration $\times$ harvest model isolated restoration and fishing effects by using a single categorical variable (rock_fish) combining restoration status (restored or unrestored) and harvest status (open or closed), with no intercept so that each combination was estimated independently. We treat this model as inferential rather than predictive and do not use its expected log predictive density (ELPD) rank to select among covariate models. From this model, we report absolute oyster densities (oysters/m²) for open and closed sites, along with the closed minus open difference (closed – open) and the open-to-closed ratio (open $\div$ closed), each with 95% CrIs, to address management questions about fishing effects in the context of restoration.

All models converged well ($R\text{-hat} \approx 1.0$), with large effective sample sizes and well-mixed chains. All data and example code used in these models are available at GitHub (https://github.com/billpine/LCR_counts).

Results

Within the Lone Cabbage locality, we sampled 15,565 m of transect at 99 unique stations (Table B2), including 1,385 m across 19 stations prerestration (periods 1–7), 1,551 m across eight stations during the pilot study (periods 10–16), and 12,629 m across 97 stations postrestoration (periods 18–26). Station totals are reported by phase, and many stations were sampled in more than one phase. Findings related to the four research questions are presented below. All densities and contrasts refer to live oysters with a shell height of 18 mm or greater. Reported densities represent three complementary summaries: (1) landscape-scale averages across all sampled sites and periods, (2) period-specific estimates (e.g., period 26), and (3) repeated-measures predictions from fixed restored stations; these are not directly comparable and are interpreted accordingly.

Research question 1: Temporal trends across the landscape

Temporal variability among periods was substantial ($SD = 0.65$; 95% CrI = 0.44–0.99), corresponding to typical differences of nearly twofold in live-oyster density ($e^{0.65} = 1.91$).

Prerestration (periods 1–7)

We observed a strong spatial gradient in oyster density, with the highest counts at inshore reefs and the lowest counts offshore. Across all reef types, densities declined through periods 1–7 and converged toward their lowest observed levels by the end of prerestration monitoring (Figure 2; Table 1). Before restoration, all offshore reefs were unrestored and are therefore shown as a single group in Figure 2. Reefs that were later

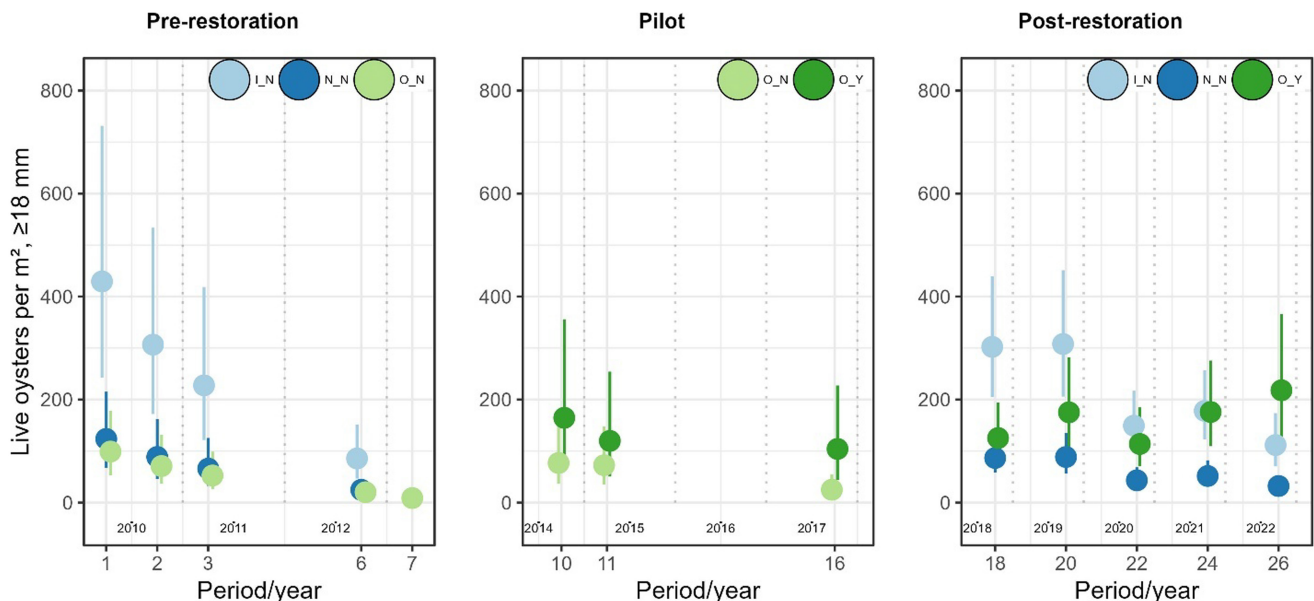


Figure 2 Trends in live-oyster density across three project phases. Points show posterior means with 95% credible intervals from a negative-binomial generalized linear mixed model (log link) with an offset of $\log(\text{transect length})$; values (oysters/m²) assume a 0.1524-m belt width. The first letter in each abbreviation defines the reef zone: I = inshore, N = nearshore, and O = offshore. The second letter defines the restoration status: N = unrestored and Y = restored with rock cultch.

Table 1 Model-based period estimates of live-oyster density (oysters/m²; ≥ 18 mm shell height) at Lone Cabbage Reef by site and restoration status. Values are predicted densities expressed as oysters per square meter. Model coefficients were estimated on a per-meter basis using an offset for log(transect length) and were converted to values per square meter for presentation.

Site	Restored?	Period	Posterior mean	95% credible interval	
				Lower	Upper
Inshore	No	1	429	243	732
Nearshore	No	1	124	67	215
Offshore	No	1	99	53	178
Inshore	No	2	307	172	534
Nearshore	No	2	89	46	162
Offshore	No	2	71	37	132
Inshore	No	3	228	121	419
Nearshore	No	3	66	32	125
Offshore	No	3	53	26	99
Inshore	No	6	86	47	152
Nearshore	No	6	25	12	46
Offshore	No	6	20	9	40
Offshore	No	7	9	4	20
Offshore	No	10	77	37	154
Offshore	Yes	10	165	73	356
Offshore	No	11	73	35	148
Offshore	Yes	11	120	51	254
Offshore	No	16	25	10	55
Offshore	Yes	16	104	44	227
Inshore	No	18	302	205	439
Nearshore	No	18	87	58	126
Offshore	Yes	18	125	81	194
Inshore	No	20	308	206	451
Nearshore	No	20	89	56	135
Offshore	Yes	20	175	105	282
Inshore	No	22	149	102	217
Nearshore	No	22	43	27	69
Offshore	Yes	22	114	70	185
Inshore	No	24	178	123	257
Nearshore	No	24	52	31	82
Offshore	Yes	24	176	110	276
Inshore	No	26	112	70	174
Nearshore	No	26	32	19	52
Offshore	Yes	26	218	128	366

restored with dolomite limestone had similarly low preres-toration densities relative to reefs that remained unrestored, supporting the comparability of the two groups before restoration.

Pilot restoration (periods 10–16)

During this phase, only offshore sites were sampled. Oyster density on restored sites rose rapidly following pilot restoration and remained consistently higher than density on unrestored offshore reefs throughout periods 10–16, even as both groups declined somewhat over this period (Figure 2; Table 1).

Full restoration (periods 18–26)

After full restoration at Lone Cabbage Reef, oyster densities rebounded strongly at offshore restored sites. Over periods 18–26, oyster densities generally declined on inshore and nearshore reefs but increased on restored offshore reefs. By period 26, restored offshore reefs supported 218 oysters/m² (95% CrI = 128–366), exceeding densities at inshore (112 oysters/m²; 95% CrI = 70–174) and nearshore (32 oysters/m²; 95% CrI = 19–52) reefs (Figure 2; Table 1). Repeated-measures estimates reflect local maximum response at fixed restored stations, whereas landscape-scale estimates average across sites and periods and therefore represent lower population-level means.

Research question 2: Repeated-measures response

The period-level random-effect SD was 0.68 (95% CrI = 0.07–1.86), consistent with variability among postrestoration periods. The wider 95% CrI reflects fewer periods and therefore a smaller n , which provides less information for the variance component in a hierarchical model, resulting in broader posteriors. At the four repeated-measures restoration stations (three transects each), posterior predictions showed a sustained and steadily increasing restoration response across all postrestoration periods (18–26). Oyster density increased from period 18 through period 26 and reached 244 oysters/m² (95% CrI = 134–433) by the end of monitoring, thus reflecting a sustained increase across all postrestoration periods (Figure 3; Table B3).

Research question 3: Spatial contrasts among reef types

Restoration of offshore reefs increased live-oyster density, especially relative to nearshore benchmarks. Averaged across all monitoring periods, restored offshore reefs supported densities comparable to those on inshore unrestored reefs and substantially higher than densities on nearshore unrestored reefs, with a ratio of approximately 3:1 (Figure 4; Table B4). Variation in transect-level densities was substantial across all groups, reflecting natural heterogeneity among reefs and periods. Notably, restored offshore reefs were the only group with no zero-count transects, suggesting that restoration not only

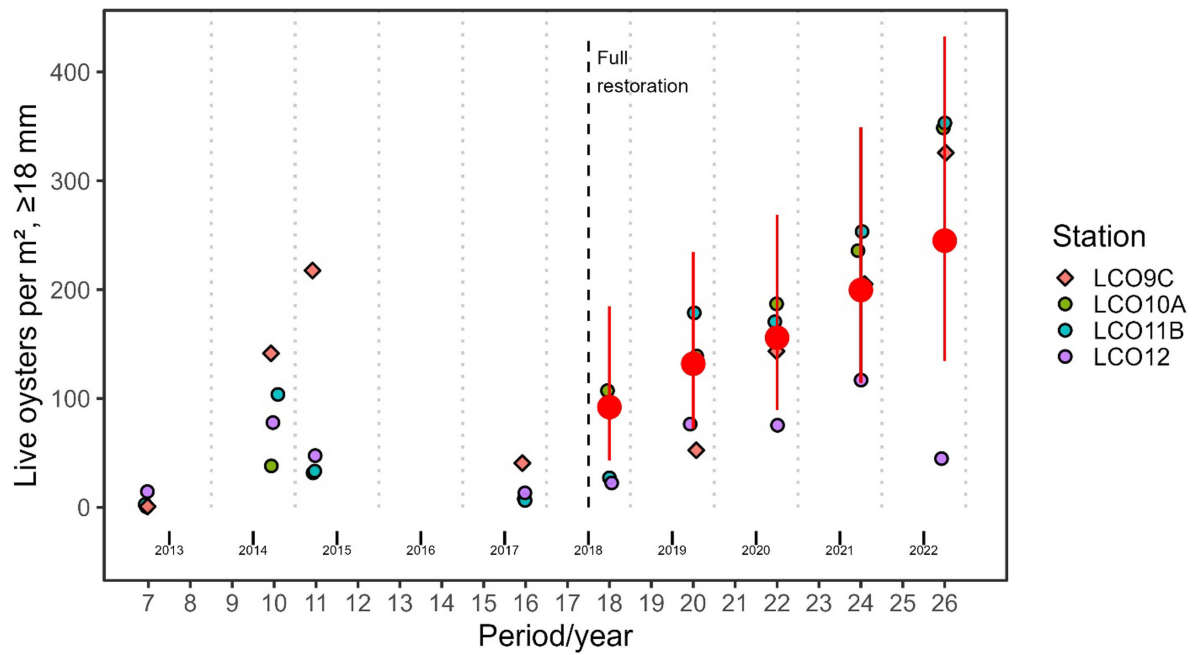


Figure 3 Repeated measures at four offshore stations on Lone Cabbage Reef that were later restored (see Figure 1), showing observed oyster densities (≥ 18 mm shell height) from 2013 to 2023. Smaller symbols (with color representing the station) show observed station–period densities, calculated as pooled live-oyster counts divided by pooled transect length and converted to square meters assuming a 0.1524-m belt width; points are jittered for visibility. Red points show posterior means ($\pm 95\%$ credible intervals) from a negative-binomial generalized linear mixed model (log link) with an offset for transect length and a period random intercept; predictions are period-specific and based on prestoration data only (2018–2023). The vertical dashed line indicates the onset of full restoration in 2018.

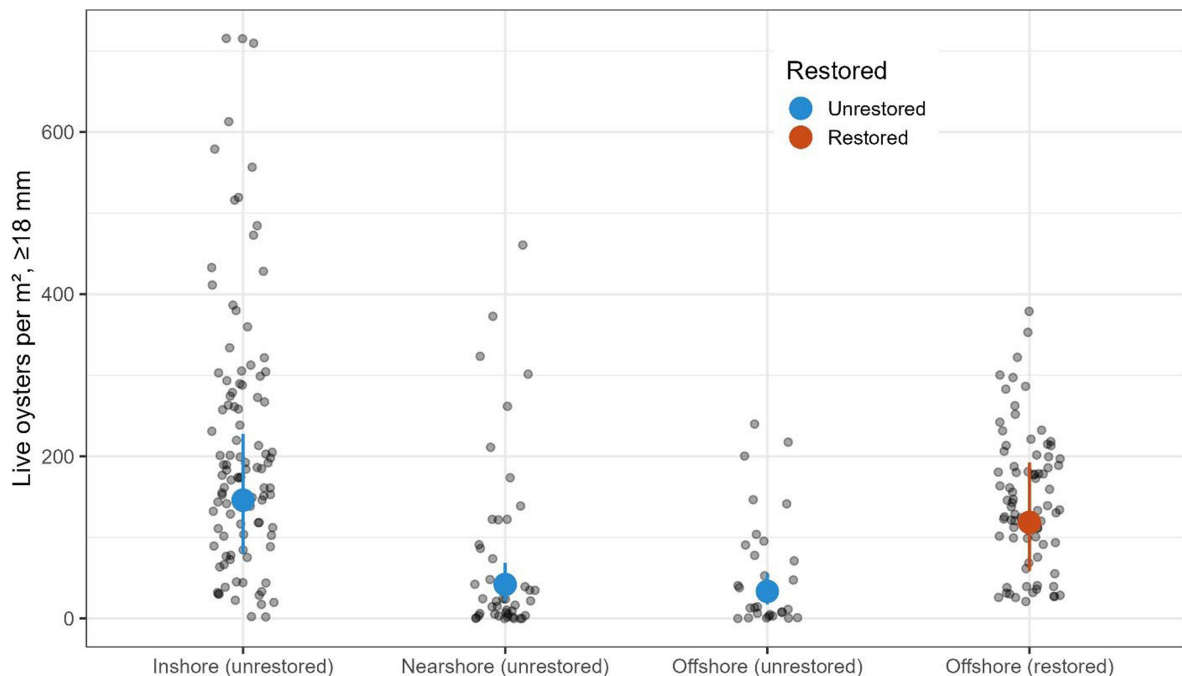


Figure 4 Live-oyster density at inshore (unrestored), nearshore (unrestored), offshore (unrestored), and offshore (restored) sites. Small black points show observed transect-level densities (overlap can appear darker). Colored points show population-average posterior mean density ($\pm 95\%$ credible intervals) from a negative-binomial generalized linear mixed model (log link) with a cell-means term for site $\times$ restoration ($0 + \text{site_rock}$), an offset for $\log(\text{transect length})$, and a period $\times$ restoration random intercept. Predictions are fixed effects only (random effects are set to zero) and thus are averaged over periods. Offshore (restored) densities are based on periods 10, 11, 16, 18, 20, 22, 24, and 26; inshore (unrestored) and nearshore (unrestored) densities are based on periods 1, 2, 3, 6, 18, 19, 20, 22, 24, and 26. See Table B2 for effort by period and Table B5 for offshore restoration/harvest summaries.

increased mean density but also eliminated the local oyster absences that occurred on all unrestored reef types. Together, these results indicate that restoration elevated offshore reefs toward inshore-like densities (i.e., inshore $\approx$ restored offshore > nearshore).

Research question 4: Covariate effects

Model ranking

We compared covariate models using PSIS-LOO on a common observation set and ranked them by ELPD. The top model from the model selection was the site $\times$ commercial landings 2-year lag specification (LandingsLag2). The next five models all used landings or trips specifications and performed similarly to each other. Discharge covariates ranked lower, and the simpler site-only and rock $\times$ fish models ranked lowest. Full model rankings and Pareto k diagnostics are provided in Table B5 and Figure 5.

Restoration contrast (offshore restored versus unrestored)

Across the top-ranked specifications, the restoration effect was large and consistent, ranging from +82 to +84 oysters/m² depending on the model (Table 2). Adding the best covariate changed little about the magnitude of the restoration gain, thus reinforcing restoration as the dominant driver of higher offshore densities.

Fishing activity (aggregate landings/trips)

Although adding fishery covariates improved the overall model fit relative to the site-only model (Δ ELPD = +14.66; SE = 6.59), the fishery effect on offshore oyster density was weak. Landings and trips specifications performed similarly to each other, and offshore coefficients on all fishery covariates were small, with 95% CrIs spanning zero in both restored and unrestored strata (Figure 5; Table B5). The improvement in overall model fit therefore reflects modest predictive gains across all sites rather than a strong or consistent directional effect of fishing activity on offshore oyster density specifically.

Simplified open versus closed to fishing on restored reefs

Point estimates suggested lower oyster density on open versus closed restored reefs, but 95% CrIs for the difference included zero and the ratio (open $\div$ closed) included 1.0, providing no clear statistical evidence that harvest status changes the inter-tidal density on restored offshore reefs (Figures 6 and 7; Table 2; Table B4). These population-average predictions are lower than the repeated-measures values in research question 2 because averaging across all postrestoration periods includes the early low-abundance periods immediately after construction.

River discharge (average or total; 0–2-year lags)

Discharge covariates did not improve out-of-sample prediction relative to the top fishery model, and the offshore

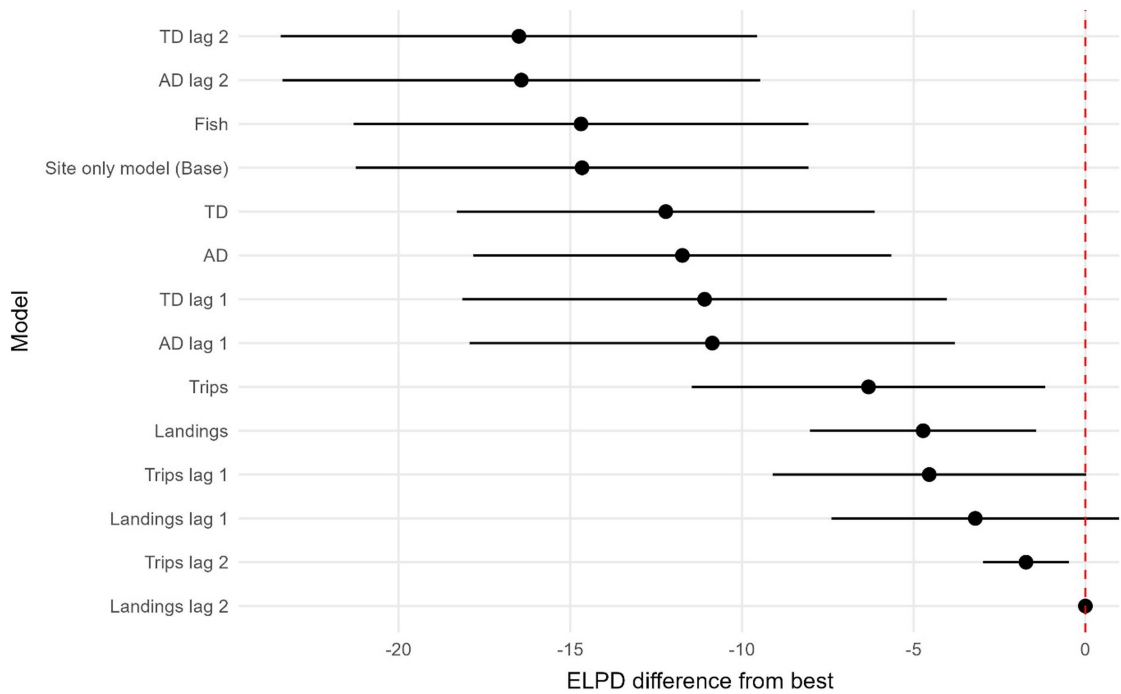


Figure 5 Model comparison for research question 4 using Pareto smoothed importance sampling leave-one-out cross validation. Points show differences (± 1 SE) in expected log predictive density (Δ ELPD) from the best-supported model; the dashed red line marks Δ ELPD = 0 (i.e., the best model). Covariate abbreviations are as follows: AD = mean daily discharge, TD = total discharge, Trips = commercial fishing trips, Landings = commercial landings, Lag1 = 1-year lag (two semiannual periods), and Lag2 = 2-year lag (four semiannual periods). Base is the site-only model (no covariate), and Fish is the site $\times$ restoration $\times$ harvest categorical model, which was used to compare restored offshore reefs that were closed versus open to harvest. Larger (less-negative) values of Δ ELPD indicate better out-of-sample predictive performance; models within approximately 1 SE of the best model are effectively tied. All models are negative-binomial generalized linear mixed models (log link) fitted to the same observations, with an offset of $\log(\text{transect length})$ and a period $\times$ restoration random intercept.

Table 2 Key effects across the four research questions. Values are posterior means with 95% credible intervals (CrIs). All generalized linear mixed models (GLMMs) are negative-binomial with a log link and include an offset for log(transect length); where applicable, models include site $\times$ covariate terms and a period $\times$ restoration random intercept. See Table B5 for offshore-specific restoration and harvest breakdowns. Abbreviations are as follows: PSIS-LOO = Pareto smoothed importance sampling leave-one-out cross validation and Δ ELPD = difference in expected log predictive density.

Research question	Effect	Definition	Oysters per m ²	
			Posterior estimate	95% CrI
1. How did live-oyster counts change across the landscape before restoration (periods 1–7), after the pilot restoration (periods 10–16), and after the full restoration (periods 18–26)? Results presented are end-of-monitoring (period 26) population-average densities.	Period-26 density, offshore restored	Population-average mean oysters per m ² at end of monitoring	218	128–366
	Period-26 density, inshore unrestored	Population-average mean oysters per m ² at end of monitoring	112	70–174
	Period-26 density, nearshore unrestored	Population-average mean oysters per m ² at end of monitoring	32	19–52
2. At the Lone Cabbage Reef restoration sites in Suwannee Sound that were sampled every year postrestoration, how did live-oyster counts change? Results presented are repeated-measures period-26 density (postrestoration), including the period random effect.	Repeated-measures postrestoration): period-26 density	Mean oysters per m ² for repeated-measures stations at end of monitoring	244	134–433
3. Across Suwannee Sound, how did average live-oyster counts compare between restored and unrestored oyster reefs while accounting for temporal and spatial variation? Results presented are population-average cell means (fixed effects only) from the cell-means negative-binomial GLMM.	Cell mean, inshore	Population-average mean oysters per m ² across periods	145	84–226
	Cell mean, nearshore	Population-average mean oysters per m ² across periods	41	23–69
	Cell mean, offshore unrestored	Population-average mean oysters per m ² across periods	33	19–56
	Cell mean, offshore restored	Population-average mean oysters per m ² across periods	113	62–192
4. Do covariates of management interest, such as Suwannee River discharge and commercial fishing activity, explain variation in oyster counts between restored and unrestored reefs while accounting for temporal variation? If so, do they explain the variation in oyster counts to a greater or lesser extent than the restoration itself? Results presented include restoration and harvest contrasts (e.g., offshore restored vs. unrestored; open vs. closed) and PSIS-LOO model comparison (Δ ELPD of best vs. site only).	Restoration contrast (offshore restored vs. unrestored)	Difference in oysters per m ²	+84	22–175
	Restoration contrast (site-only model, no covariates)	Difference in oysters per m ²	+82	26–157
	Harvest effect (closed to harvest vs. open to harvest, offshore restored)	Difference in oysters per m ²	29	–5 to 78
	Model comparison	Δ ELPD(best vs. site only)	+14.66	NA

restoration contrast remained essentially unchanged across all discharge specifications (Figure 5; Table B5). Discharge slopes were generally small and uncertain once site and period were included, with only the 1-year lag (Lag1) terms showing a positive association on offshore unrestored reefs; all offshore restored slopes overlapped zero for every discharge specification.

Covariate slopes

Discharge and fishery covariate slopes were consistently weak and uncertain across reef types and lag structures, with 95% CrIs spanning zero in nearly all cases (Figure 5; Table B5). Together, these results reinforce that restoration status is the dominant driver of offshore oyster density, with discharge and fishery proxies offering limited additional explanatory power.

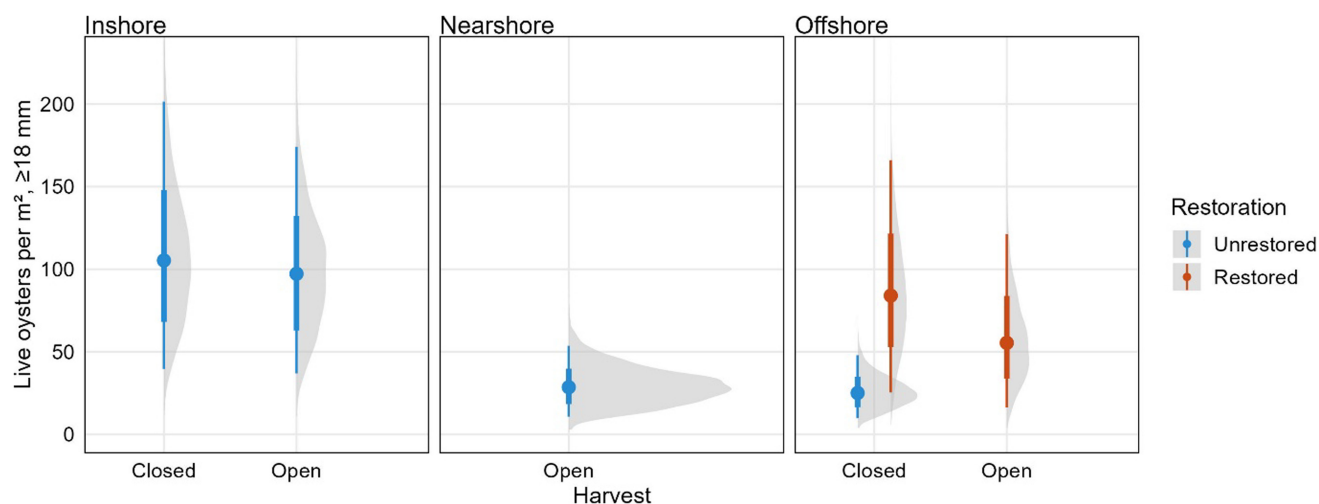


Figure 6 Predicted live-oyster density (oysters/m²; ≥18 mm shell height) by harvest status (open or closed) and restoration (restored or unrestored) across reef zones. Points show population-average posterior means with 95% credible intervals from a negative-binomial generalized linear mixed model (log link) with an offset of log(transect length), a cell-means fixed effect for site × restoration × harvest, and a period × restoration random intercept. Predictions are fixed effects only (random effects are set to zero), so values are averaged over periods. Densities per square meter were obtained by converting from per-meter transect predictions using a transect width of 0.1524 m. Reef category predictions are based on the following sampled periods: periods 1, 2, 3, 6, 18, 19, 20, 22, 24, and 26 for inshore unrestored (open/closed) and nearshore unrestored (open); periods 1, 2, 3, 6, 7, 10, 11, and 16 for offshore unrestored (closed); periods 10, 11, 16, 18, 20, 22, 24, and 26 for offshore restored (closed); and periods 18, 20, 22, 24, and 26 for offshore restored (open). See [Table B2](#) for sampling effort by period and [Table B5](#) for summary statistics.

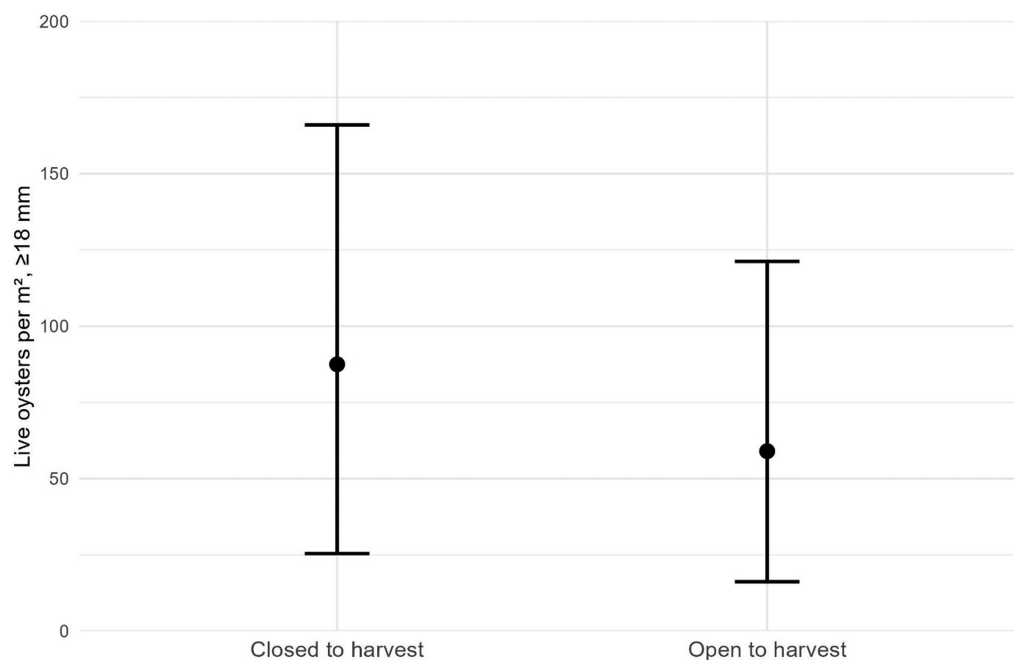


Figure 7 Live-oyster density (oysters/m²; ≥18 mm shell height) on restored offshore reefs that were closed versus open to harvest. Points show population-average posterior mean densities (±95% credible intervals) from a negative-binomial generalized linear mixed model (log link) with an offset of log(transect length) and a categorical factor distinguishing restored reefs that were open or closed to harvest. Predictions are fixed effects only (random effects are set to zero), so values are averaged over periods. Estimates for both categories included periods 18, 20, 22, 24, and 26; estimates for the closed category also included periods 10, 11, and 16. See [Table B2](#) for effort by period and [Table B5](#) for offshore restoration/harvest summaries.

Discussion

Restoration at Lone Cabbage Reef produced a large, sustained increase in oyster density. By 5 years postrestoration, restored offshore sites averaged 218 oysters/m² (95% CrI = 128–366),

roughly 24 times higher than the prerestoration low of 9 oysters/m² and greater than densities at inshore (112 oysters/m²; 95% CrI = 70–174) and nearshore (32 oysters/m²; 95% CrI = 19–52) reefs. These gains were attributable to

restoration rather than to systemwide recovery (Figures 2, 3, and 6; Tables 1 and 2). Both landscape comparisons and repeated-measures stations showed the same pattern: Oyster density at restored offshore reefs increased from near zero and remained elevated, while density at unrestored reefs declined or stayed low. This strong, sustained response contrasts with the preceding decade, during which offshore reefs did not recover on their own (Frederick et al., 2016; Moore et al., 2020). After restoration and period-to-period variation were accounted for, discharge and fishery covariates added little explanatory or predictive power. In practical terms, the sequence of management actions matters for oyster resources in Suwannee Sound. Our results suggest that the first step would be to restore degraded reefs with suitable, durable cultch to a suitable elevation to re-establish a resilient, self-sustaining oyster population; after that, expectations about other covariates can be calibrated. Where harvest occurs, managing cultch is essential. In this system, restoration was the primary driver of recovery because it supplied the missing substrate. Adding rock at persistent elevation rebuilt settlement area, likely improved larval retention and survival, and re-established positive feedbacks that support oyster recruitment and persistence. Consistent with this substrate-driven recovery mechanism, restored offshore reefs' oyster density increased from near zero immediately after construction to their highest observed densities within a few years, while unrestored reefs continued to decline (see Figures 2 and 4; Table 1). Prioritizing restoration and then conserving cultch likely represent the key pathway to resilient oyster populations in Suwannee Sound.

Mechanistically, the geomorphic context likely matters. Each reef chain in Suwannee Sound probably formed under different sea levels, and offshore reefs at Lone Cabbage are most likely relict shorelines (Grinnell, 1972; Hine et al., 1988; Wright et al., 2005). Reef elevation is a critical determinant of restoration success. Lenihan and Peterson (1998) demonstrated that reduced reef height amplifies the effects of hypoxia and physical stress on oyster survival. Schulte et al. (2009) showed that high-relief reefs in Chesapeake Bay supported oyster densities approximately four times greater than low-relief reefs, with the high-relief reefs having stronger recruitment and reduced sedimentation. Furthermore, Colden et al. (2017) demonstrated that reef height can operate as a threshold driver of restored oyster reef dynamics, with reefs below a critical elevation failing to sustain populations regardless of larval supply. Our intervention added durable dolomite limestone, replacing degraded cultch and increasing the reef elevation to levels similar to those of inshore areas (Aufmuth et al., 2025); therefore, decades to centuries of natural cultch accumulation were effectively compressed into a single restoration event (Ridge et al., 2015, 2017). By restoring substrate and reef elevation, our restoration may have overcome hysteresis associated with cultch loss (Johnson et al., 2022; Walker et al., 2012), enabling persistent oyster recruitment and population growth. This suggests that our restoration strategy would work at similar offshore, inshore, or nearshore sites with the correct target elevation, appropriate environmental conditions, and adequate larval supply. The abundance gains documented here are reinforced by

complementary evidence that restoration changed reef structure more broadly. A concurrent analysis of more than 27,000 oysters across the region found that restored offshore sites achieved mean shell heights comparable to those at inshore reefs within 2 years of restoration, including in areas open to harvest (Pine et al., 2026). Although oyster size was not analyzed in the present study, these results indicate that restored offshore reefs supported size structures similar to those observed for inshore reefs, including oysters approaching or exceeding harvestable size, which suggests that the higher densities observed here reflect sustained population recovery rather than transient recruitment pulses. Aufmuth et al. (2025) likewise documented that restored Lone Cabbage Reef was built to match the elevation of healthy inshore reefs and that this elevation was retained over multiple years, including after major storm events. Together, the convergence of higher abundance in this study with independent evidence of improved size structure and sustained reef elevation suggests that restoration moved offshore reefs toward a more persistent, higher-productivity state similar to inshore reference reefs rather than their prerecovery condition. This convergence in elevation, oyster density, and shell height suggests that restoration success was driven not only by adding cultch but also by restoring reefs with suitable material to appropriate intertidal elevations, which promotes larval retention, reef persistence, and positive feedbacks (Aufmuth et al., 2025; Bost et al., 2021). Field assessments of relationships between cultch mass and live-oyster density (Casteel et al., 2025) further support threshold dynamics, showing a saturating response with a potential tipping point at low cultch levels. Taken together, these findings underscore the need to integrate biological metrics, such as oyster density and size structure; structural metrics, such as cultch mass and type; and geomorphic metrics, such as elevation, into restoration planning and monitoring (Colden et al., 2017; Goelz et al., 2020; Lipcius et al., 2021). The restoration response at Lone Cabbage Reef suggests that when elevation and cultch thresholds are met, degraded offshore reefs can support oyster populations that are comparable to or that exceed inshore benchmarks within a few years, despite their historical decline (Figure B5).

River discharge and oyster populations: Insights from long-term data

Oyster reef restoration had a much larger effect on oyster density than any influence of river discharge observed in this study (research question 4). River discharge remains a major conservation concern in Florida (Florida v. Georgia, 2021; Kelly, 2019), and Moore et al. (2020) previously reported a positive 1-year lag relationship between mean daily discharge and intertidal oyster counts at Lone Cabbage Reef using data from periods 1–17. Extending the record through period 26 added wetter postrestoration years and denser sampling at Lone Cabbage Reef, providing a broader test of that relationship. Differences in analytical approach, together with continued loss of remnant cultch and live oysters on unrestored reefs, likely explain why the longer record showed weaker and less consistent discharge effects. More broadly, our results suggest

that without sufficient cultch, discharge effects on intertidal oyster density are small and uncertain: Restoration provides the structural capacity for oysters to persist, whereas hydrology modulates populations around that capacity.

Our results also suggest that the weak discharge signal was not simply an artifact of model structure. Because discharge was aggregated at the same temporal scale as the period random effect, we refitted the top discharge model (site $\times$ total discharge, 1-year lag) using fixed period effects and compared it with the original random-intercept model using PSIS-LOO. The fixed-effects model did not improve predictive performance ($\Delta\text{ELPD} = -5.6 \pm 33.2$), and slope estimates were similar between specifications. Under the fixed-effects model, the offshore unrestored slope for Lag1 total discharge was positive (mean = 0.79; 95% CrI = 0.29–1.28), whereas the offshore restored slope was smaller and overlapped zero (mean = 0.25; 95% CrI = -0.05 to 0.57). These results support the primary specification with a period random intercept.

Using the top PSIS-LOO discharge model with a 1-year lag, a 1-SD increase in discharge relative to the 2010–2023 mean at the U.S. Geological Survey Wilcox gauge (mean = 275 m³/s [9.715 ft³/s]; SD = 125 m/s [4,410 ft³/s]) corresponded to predicted increases of +39 oysters/m² (95% CrI = 8–96) on offshore unrestored reefs and +22 oysters/m² (95% CrI = -34 to 93) on offshore restored reefs. Thus, higher-discharge years were associated with a larger increase on unrestored reefs and a smaller, uncertain increase on restored reefs. Because a 1-SD increase represents a large change in discharge, these predictions should be interpreted as illustrative rather than implying a linear response across smaller discharge changes. This pattern is consistent with the idea that remnant substrate on degraded reefs can still support transient recruitment pulses under favorable conditions, even if such pulses are not sufficient to drive sustained recovery.

Together, these results support a coherent picture: Restoration provides the structural capacity for high density, and hydrology modulates that capacity. Higher-discharge years may temporarily narrow the restored–unrestored oyster density gap where remnant substrate persists, but discharge does not substitute for cultch; higher-discharge year gains on unrestored reefs are uncertain and unlikely to persist without rebuilding reef structure (Aufmuth et al., 2025; Colden et al., 2017; Pine et al., 2015; Solinger et al., 2022).

Implications under changing hydroclimate

In the southeastern United States, discharge variability is expected to increase in the future alongside greater withdrawals of surface water and subsurface water (Nieto Ferreira et al., 2018; Sun, 2013), potentially driving more frequent low-flow periods and altered salinity/nutrient regimes in Suwannee Sound (Marshall et al., 2021; Sinnickson et al., 2021). If remnant cultch continues to decline, episodic recruitment gains should decrease, lowering live-oyster and shell accrual and risking thresholds/tipping points (Casteel et al., 2025; Johnson et al., 2022). Environmental conditions can boost recruitment in some years, but favorable conditions do not replace substrate: Without cultch, few oysters are present to respond (see research question 1; Casteel et al., 2025).

This likely helps to explain why discharge–oyster relationships vary among systems (Fisch & Pine, 2016; Gledhill et al., 2020; Kelly, 2019; Pine et al., 2015), why some restoration responses are transient (Pine et al., 2023), and why the brief inshore increase that we observed in periods 18–19 did not persist.

Fishing effects on intertidal reefs

Similar to river discharge, restoration had a much larger effect on oyster density than the range of commercial harvest observed in recent decades (research question 4). Under closed harvest offshore, restored reefs had higher oyster density than unrestored reefs (ratio = 3.71, 95% CrI = 1.18–8.39; Figure 6), corresponding to an absolute restoration gain of approximately 82–84 oysters/m² across models (Table 2; Table B4). Even at intertidal sites that are open to harvest, predicted densities of approximately 59 oysters/m² translate to fewer than one legal oyster per square meter (~0.89), given an observed legal-size fraction of about 1.5% (Pine et al., 2026; Figure 7; Table B4). Together, these results indicate that harvest effects on intertidal oyster reefs are small to uncertain, whereas restoration produces large proportional and absolute increases. This pattern suggests that cultch provides the structural capacity for reefs to persist, while fishing primarily modulates populations around that capacity.

This does not imply that harvest cannot affect oyster populations, but several factors likely weaken the closure signal on intertidal reefs. Intertidal oysters rarely reach legal size because growth is limited by air exposure, and only about 1.5% of more than 27,000 intertidal oysters measured in this region were legal size (Pine et al., 2026). Larger oysters are more common on adjacent subtidal bars targeted by the fishery, suggesting a source–sink dynamic in which subtidal reefs supply legal-size oysters to nearby intertidal areas (Moore et al., 2020). Anecdotal reports from fishers, including hand-collection of legal oysters from deeper areas near restored Lone Cabbage Reef, are consistent with this interpretation.

Such a source–sink dynamic might seem to imply that subtidal harvest could reduce intertidal abundance by removing potential settlers, but the evidence does not support this conclusion. Closing the subtidal fishery is therefore unlikely to reverse intertidal declines based on current understanding of oyster population dynamics in this area. There is no fishery-independent time series for oyster counts on subtidal reefs in this area. Given the strong response to substrate restoration and evidence for a tipping point in the cultch–live-oyster relationship (Casteel et al., 2025), intertidal oyster declines are more likely linked to loss of cultch on intertidal reefs. Removals of live oysters and shell from subtidal bars are unlikely to influence intertidal cultch. An understanding of recruitment and shell dynamics across intertidal and subtidal habitats remains critical for effective management (Camp et al., 2026; Johnson et al., 2022; Solinger et al., 2022).

The data from Suwannee Sound support a simple message: Intertidal reefs can be successfully restored even where harvest is allowed, and harvest policies are best framed to sustain cultch and reef structure as the prerequisites for sustainable

harvest rather than as tools to protect larval production alone. This remains an important area for future work.

Study limitations

We did not model oyster size explicitly. Transect counts excluded oysters smaller than 18 mm, so our density estimates are conservative. A companion analysis for Suwannee Sound indicated that restored reefs also shift size distributions toward larger individuals, consistent with substrate addition improving multiple population metrics (Pine et al., 2026). Because the reefs that we monitored are intertidal, growth remains primarily below the legal harvest threshold. Thus, our results should not be interpreted as increases in harvestable biomass. Overall, while our models focus on counts, restoration effects likely extend to size structure (Pine et al., 2026).

A second limitation is that we modeled river discharge and fishery covariates as common effects across restored offshore reefs rather than reef-specific responses. A per-reef approach (e.g., tracking salinity or fishing at each reef) could be informative but risks pseudoreplication and is logistically costly given the number of restored sites and the random selection of unrestored reefs each year. We also did not model nonlinear relationships or complex lags between covariates and oyster density. Finally, there is a spatial scale mismatch between regional, largely subtidal fishery indices (total trips and landings) and intertidal transect density. Addressing these issues will require paired intertidal–subtidal monitoring, finer maps of subtidal effort and landings, and outcome metrics beyond density.

Third, our results come from a single restoration project in one estuary. Although we sampled repeatedly over time and space, spatial replication was constrained by the project's fixed location, so findings reflect Lone Cabbage Reef and likely similar offshore habitats in Suwannee Sound. That said, our models estimate contrasts between restored and unrestored reefs while accounting for site type and time, and we had both before/after sampling at restored offshore reefs and postrestoration comparisons with unrestored inshore and nearshore reefs. Broader generalization will require additional replication across estuaries, but consistent patterns across models, together with random effects for time or site, support the robustness of the restoration effects observed here.

More broadly, our study highlights a common challenge in coastal restoration: turning context-specific projects into transferable knowledge. Pine et al. (2022) suggested a nested adaptive management framework in which individual projects serve as learning platforms within a larger program. Local outcomes inform system-level hypotheses, and restoration is iteratively refined and scaled through single- and double-loop learning. Viewed in this light, the Lone Cabbage Reef project is not just a stand-alone success but also represents a step toward generalizable guidance on when and where substrate addition yields ecological recovery and how this recovery fits into management actions, such as fishery closures.

Insights and future directions

Substrate addition using appropriate cultch type, size, and elevation can reverse local oyster declines in substrate-limited

systems with adequate larval supply, as demonstrated here (but see contrasting results in six Florida studies; Pine et al., 2023). Achieving recovery at landscape scales, however, requires sustained natural recruitment supported by adequate cultch (Casteel et al., 2025; Pine et al., 2015). Long-term success therefore depends on understanding both the ecological drivers of cultch loss and the social and institutional context of restoration decisions. Iterative experiments embedded within a broader program of monitoring, modeling, and stakeholder engagement can accelerate learning and improve restoration outcomes (Johnson et al., 2023; Pine et al., 2022). This approach, including single- and double-loop learning, shifts restoration from one-off actions to a long-term, evolving strategy. However, learning reduces uncertainty only if it influences the decisions that govern real projects.

Because agencies typically set restoration objectives, designs, and success criteria, a failure to institutionalize lessons means that management does not change and uncertainty persists. The Lone Cabbage Reef project shows the value of adaptive restoration and how well-monitored, spatially explicit interventions can produce quantifiable outcomes that inform future decisions (Appendix C). It also underscores restoration's limits when used alone. Without parallel efforts to diagnose and address systemic drivers, such as cultch loss and altered recruitment in a changing climate, restoration cannot fully reverse long-term oyster reef decline (Pine et al., 2015).

Restoration fits within a broader adaptive program that integrates field experiments, predictive modeling, and structured decision making for estuarywide oyster management. At Lone Cabbage Reef, oyster densities roughly tripled within 5 years, providing a benchmark for what targeted, well-designed interventions can achieve. Sustaining and expanding these gains will require long-term investment, refinement of restoration methods, collaborative governance, and harvest policies that conserve cultch. The Lone Cabbage Reef results demonstrate that restoration can rapidly rebuild oyster populations, but durable recovery will depend on restoring the substrate dynamics that sustain reefs over decades.

Data availability

All data and example code are available at GitHub (https://github.com/billpine/LCR_counts).

Ethics statement

The research meets the ethical guidelines and legal requirements for observational studies in the United States.

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Conflicts of Interest

The authors declare no conflicts of interest.

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References

- Aufmuth, J., Moore, J. F., Pine, W., III, Frederick, P., & Ennis, B. (2025). Intertidal oyster reef elevation restoration: Using a durable substrate on Lone Cabbage Reef, Florida, United States. *Restoration Ecology*, 33, Article e70164. <https://doi.org/10.1111/rec.70164>
- Baggett, L. P., Powers, S. P., Brumbaugh, R. D., Coen, L. D., DeAngelis, B. M., Greene, J. K., Hancock, B. T., Morlock, S. M., Allen, B. L., Breitburg, D. L., Bushek, D., Grabowski, J. H., Grizzle, R. E., Grosholz, E. D., La Peyre, M. K., Luckenbach, M. W., McGraw, K. A., Piehler, M. F., Westby, S. R., & Zu Ermgassen, P. S. E. (2015). Guidelines for evaluating performance of oyster habitat restoration. *Restoration Ecology*, 23, 737–745. <https://doi.org/10.1111/rec.12262>
- Barbour, T. E., Sassaman, K. E., Almeyda Zambrano, A. M., Broadbent, E. N., Wilkinson, B., & Kanaski, R. (2019). Rare pre-Columbian settlement on the Florida Gulf coast revealed through high-resolution drone LiDAR. *Proceedings of the National Academy of Sciences of the United States of America*, 116, 23493–23498. <https://doi.org/10.1073/pnas.1911285116>
- Barnett, C. (2021, April 9). Why America's water wars are futile. *Tampa Bay Times*. <https://www.tampabay.com/opinion/2021/04/09/why-americas-water-wars-are-futile-column/>
- Beisner, B., Haydon, D., & Cuddington, K. (2003). Alternative stable states in ecology. *Frontiers in Ecology and the Environment*, 1, 376–382. [https://doi.org/10.1890/1540-9295\(2003\)001\[0376:ASSIE\]2.0.CO;2](https://doi.org/10.1890/1540-9295(2003)001[0376:ASSIE]2.0.CO;2)
- Bendick, R., DeAngelis, B., & Blitch, S. (2018). *Oyster restoration in the Gulf of America: Proposals from The Nature Conservancy*. The Nature Conservancy. <https://www.nature.org/content/dam/tnc/nature/en/documents/gulf-of-mexico-oyster-restoration-plan.pdf>
- Bersosa Hernández, A., Brumbaugh, R. D., Frederick, P., Grizzle, R., Luckenbach, M. W., Peterson, C. H., & Angelini, C. (2018). Restoring the Eastern Oyster: How much progress has been made in 53 years? *Frontiers in Ecology and the Environment*, 16, 463–471. <https://doi.org/10.1002/fee.1935>
- Blomberg, B. N., Pollack, J. B., Montagna, P. A., & Yoskowitz, D. W. (2018). Evaluating the US Estuary Restoration Act to inform restoration policy implementation: A case study focusing on oyster reef projects. *Marine Policy*, 91, 161–166. <https://doi.org/10.1016/j.marpol.2018.02.014>
- Bost, M. C., Rodriguez, A. B., Ridge, J. T., Miller, C. B., & Fegley, S. R. (2021). Natural intertidal oyster reef growth across two landscape settings and tidal ranges. *Estuaries and Coasts*, 44, 2118–2131. <https://doi.org/10.1007/s12237-021-00925-2>
- Bürkner, P. C. (2017). brms: An R package for Bayesian multi-level models using Stan. *Journal of Statistical Software*, 80, 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Bürkner, P. C. (2021). Bayesian item response modeling in R with brms and Stan. *Journal of Statistical Software*, 100, 1–54. <https://doi.org/10.18637/jss.v100.i05>
- Camp, E. V., Pine, W., Casola, W. R., Benjamin, E. D., Palmer, C., Klizentyte, K., & Lorenzen, K. (2026). Implications of depensation on oyster recruitment, population collapse, restoration, and sustainable fisheries management. *Fisheries Research*, 293, Article 107627. <https://doi.org/10.1016/j.fishres.2025.107627>
- Camp, E. V., Pine, W. E., III, Havens, K., Kane, A. S., Walters, C. J., Irani, T., Lindsey, A. B., & Morris, J. G., Jr. (2015). Collapse of a historic oyster fishery: Diagnosing causes and identifying paths toward increased resilience. *Ecology and Society*, 20, Article 45. <https://doi.org/10.5751/ES-07821-200345>
- Casteel, J. L., Pine, W. E., Fisch, N. C., Moore, J. F., Richardson, M. S., Ahrens, R. N. M., Camp, E. V., Cahill, C. L., & Walters, C. J. (2025). Oyster cultch-recruit patterns provide new insight into the restoration and management of a critical resource. *Marine and Coastal Fisheries: Dynamics, Management, and Ecosystem Science*, 17, Article vtaf018. <https://doi.org/10.1093/mcfafs/vtaf018>
- Coen, L. D., Brumbaugh, R. D., Bushek, D., Grizzle, R., Luckenbach, M. W., Posey, M. H., Powers, S. P., & Tolley, S. G. (2007). Ecosystem services related to oyster restoration. *Marine Ecology Progress Series*, 341, 303–307. <https://doi.org/10.3354/meps341303>
- Coen, L. D., & Luckenbach, M. W. (2000). Developing success criteria and goals for evaluating oyster reef restoration: Ecological function or resource exploitation? *Ecological Engineering*, 15, 323–343. [https://doi.org/10.1016/S0925-8574\(00\)00084-7](https://doi.org/10.1016/S0925-8574(00)00084-7)
- Colden, A. M., Latour, R. J., & Lipcius, R. N. (2017). Reef height drives threshold dynamics of restored oyster reefs. *Marine Ecology Progress Series*, 582, 1–13. <https://doi.org/10.3354/meps12362>
- Deepwater Horizon Natural Resource Damage Assessment Trustees. (2017). *Deepwater Horizon oil spill: Final programmatic damage assessment and restoration plan and final programmatic environmental impact statement*. National Oceanic and Atmospheric Administration. <https://www.doi.gov/deepwaterhorizon/programmatic-damage-assessment-and-restoration-planprogrammatic-environmental>

- Du, J., Park, K., Jensen, C., Dellapenna, T. M., Zhang, W. G., & Shi, Y. (2021). Massive oyster kill in Galveston Bay caused by prolonged low-salinity exposure after Hurricane Harvey. *The Science of the Total Environment*, 774, Article 145132. <https://doi.org/10.1016/j.scitotenv.2021.145132>
- Eleuterius, L. N., & Otvos, E. G., Jr. (1979). Floristic and geologic aspects of Indian middens in salt marshes of Hancock County, Mississippi. *SIDA, Contributions to Botany*, 8, 102–112.
- Fisch, N. C., & Pine, W. E. (2016). A complex relationship between freshwater discharge and oyster fishery catch per unit effort in Apalachicola Bay, Florida: An evaluation from 1960 to 2013. *Journal of Shellfish Research*, 35, 809–825. <https://doi.org/10.2983/035.035.0409>
- Florida Fish and Wildlife Conservation Commission. (2025). *Commercial fisheries landings in Florida* [Online database]. Florida Fish and Wildlife Research Institute. Retrieved August 29, 2025, from <https://myfwc.com/research/salt-water/fishstats/commercial-fisheries/landings-in-florida/>
- Florida v. Georgia, 585 U.S. _ (docket no. 42, slip op.) (2021).
- Frederick, P., Vitale, N., Pine, B., Seavey, J., & Sturmer, L. (2016). Reversing a rapid decline in oyster reefs: Effects of durable substrate on oyster populations, elevations, and aquatic bird community composition. *Journal of Shellfish Research*, 35, 359–367. <https://doi.org/10.2983/035.035.0210>
- Gabry, J., Češnovar, R., Johnson, A., & Bronder, S. (2025). *cmdstanr: R interface to 'CmdStan'* (R package version 0.9.0) [Computer software]. GitHub. <https://github.com/stan-dev/cmdstanr>
- Gledhill, J. H., Barnett, A. F., Slattery, M., Willett, K. L., Easson, G. L., Otts, S. S., & Gochfeld, D. J. (2020). Mass mortality of the Eastern Oyster *Crassostrea virginica* in the western Mississippi Sound following unprecedented Mississippi River flooding in 2019. *Journal of Shellfish Research*, 39, 235–244. <https://doi.org/10.2983/035.039.0205>
- Goelz, T., Vogt, B., & Hartley, T. (2020). Alternative substrates used for oyster reef restoration: A review. *Journal of Shellfish Research*, 39, 1–12. <https://doi.org/10.2983/035.039.0101>
- Grabowski, J. H. (2004). Habitat complexity disrupts predator–prey interactions but not the trophic cascade on oyster reefs. *Ecology*, 85, 995–1004. <https://doi.org/10.1890/03-0067>
- Grabowski, J. H., Brumbaugh, R. D., Conrad, R. F., Keeler, A. G., Opaluch, J. J., Peterson, C. H., Piehler, M. F., Powers, S. P., & Smyth, A. R. (2012). Economic valuation of ecosystem services provided by oyster reefs. *Bioscience*, 62, 900–909. <https://doi.org/10.1525/bio.2012.62.10.10>
- Grinnell, R. S., Jr. (1972). *Structure and development of oyster reefs on the Suwannee River delta, Florida* [Doctoral dissertation, State University of New York]. Graduate Dissertations and Theses. https://orb.binghamton.edu/dissertation_and_theses/163.
- Grizzle, R., Ward, K., Konisky, R., Greene, J., Abeels, H., & Atwood, R. (2021). Oyster reef restoration in New Hampshire, USA: Lessons learned during two decades of practice. *Ecological Restoration*, 39, 260–273. <https://doi.org/10.3368/er.39.4.260>
- Hine, A. C., Belknap, D. F., Hutton, J. G., Osling, E. B., & Evans, M. W. (1988). Recent geological history and modern sedimentary processes along an incipient, low-energy, epicontinental-sea coastline: Northwest Florida. *Journal of Sedimentary Petrology*, 58, 567–579. <https://doi.org/10.1306/212F8DF5-2B24-11D7-8648000102C1865D>
- Hintenlang, L. L., Brooks, R. M., & Kane, A. S. (2023). Assessing cumulative stressors, state shift, and the current outlook for oyster habitat in Apalachicola Bay, Florida. *Journal of Shellfish Research*, 42, 479–489. <https://doi.org/10.2983/035.042.0311>
- Holmes, N. H., & Trickey, E. B. (1974). Late Holocene sea-level oscillations in Mobile Bay. *American Antiquity*, 39, 122–124. <https://doi.org/10.2307/279226>
- Howie, A. H., & Bishop, M. J. (2021). Contemporary oyster reef restoration: Responding to a changing world. *Frontiers in Ecology and Evolution*, 9, Article 689915. <https://doi.org/10.3389/fevo.2021.689915>
- Hutchings, J. A. (1995). Seasonal marine protected areas within the context of spatio-temporal variation in the Northern Cod fishery. In N. L. Shackell & J. H. M. Willison (Eds.), *Marine protected areas and sustainable fisheries: Proceedings of the symposium on marine protected areas and sustainable fisheries conducted at the Second International Conference on Science and the Management of Protected Areas, held at Dalhousie University, Halifax, Nova Scotia, Canada, 16–20 May 1994* (pp. 39–47). Science and Management of Protected Areas Association.
- Johnson, F. A., Camp, E. V., Gandy, R., & Pine, W. E., III. (2023). Demography of oysters pre- and postcollapse in Apalachicola Bay, Florida, using stage-based counts. *Marine and Coastal Fisheries: Dynamics, Management, and Ecosystem Science*, 15, Article e10244. <https://doi.org/10.1002/mcf2.10244>
- Johnson, F. A., Pine, W. E., III, & Camp, E. V. (2022). A cautionary tale: Management implications of critical transitions in oyster fisheries. *Canadian Journal of Fisheries and Aquatic Sciences*, 79, 1269–1281. <https://doi.org/10.1139/cjfas-2021-0133>
- Kelly, P. J. (2019). *State of Florida, plaintiff v. State of Georgia, defendant, report of the special master*. Supreme Court of the United States. https://www.supremecourt.gov/Docket-PDF/22/220142/125423/20191213102743287_142%20Original%20SM%20Report.pdf
- Kopecky, K. L., Stier, A. C., Schmitt, R. J., Holbrook, S. J., & Moeller, H. V. (2023). Material legacies can degrade resilience: Structure-retaining disturbances promote regime shifts on coral reefs. *Ecology*, 104, Article e4006. <https://doi.org/10.1002/ecy.4006>
- La Peyre, M. K., Humphries, A. T., Casas, S. M., & La Peyre, J. F. (2014). Temporal variation in development of ecosystem services from oyster reef restoration. *Ecological Engineering*, 63, 34–44. <https://doi.org/10.1016/j.ecoleng.2013.12.001>
- La Peyre, M. K., Marshall, D. A., Buie, S. C. L., Hijuelos, A., & Steyer, G. D. (2022). Are we falling short on restoring oysters at a regional scale? *Environmental Management*, 70, 581–592. <https://doi.org/10.1007/s00267-022-01691-y>
- Lee, A. (2023, March 22). Mississippi has invested millions of dollars to save its oysters. They're disappearing anyway.

- ProPublica. <https://www.propublica.org/article/mississippi-spent-millions-failed-to-save-oyster-reefs>
- Lenihan, H. S. (1999). Physical–biological coupling on oyster reefs: How habitat structure influences individual performance. *Ecological Monographs*, 69, 251–275. [https://doi.org/10.1890/0012-9615\(1999\)069\[0251:PCO-OR\]2.0.CO;2](https://doi.org/10.1890/0012-9615(1999)069[0251:PCO-OR]2.0.CO;2)
- Lenihan, H. S., & Peterson, C. H. (1998). How habitat degradation through fishery disturbance enhances impacts of hypoxia on oyster reefs. *Ecological Applications*, 8, 128–140. [https://doi.org/10.1890/1051-0761\(1998\)008\[0128:HH-DTFD\]2.0.CO;2](https://doi.org/10.1890/1051-0761(1998)008[0128:HH-DTFD]2.0.CO;2)
- Lipcius, R. N., Zhang, Y., Zhou, J., Shaw, L. B., & Shi, J. (2021). Modeling oyster reef restoration: Larval supply and reef geometry jointly determine population resilience and performance. *Frontiers in Marine Science*, 8, Article 677640. <https://doi.org/10.3389/fmars.2021.677640>
- Marshall, D. A., La Peyre, M. K., Palmer, T. A., Guillou, G., Sterba-Boatwright, B. D., Beseres Pollack, J., & Lebreton, B. (2021). Freshwater inflow and responses from estuaries across a climatic gradient: An assessment of northwestern Gulf of Mexico estuaries based on stable isotopes. *Limnology and Oceanography*, 66, 3568–3581. <https://doi.org/10.1002/lno.11899>
- Mattson, R. A. (2002). A resource-based framework for establishing freshwater inflow requirements for the Suwannee River estuary. *Estuaries*, 25, 1333–1342. <https://doi.org/10.1007/BF02692228>
- Moore, J. F., & Pine, W. E., III. (2021). Bootstrap methods can help evaluate monitoring program performance to inform restoration as part of an adaptive management program. *PeerJ*, 9, Article e11378. <https://doi.org/10.7717/peerj.11378>
- Moore, J. F., Pine, W. E., III, Frederick, P. C., Beck, S., Moreno, M., Dodrill, M. J., Boone, M., Sturmer, L., & Yurek, S. (2020). Trends in oyster populations in the northeastern Gulf of Mexico: An assessment of river discharge and fishing effects over time and space. *Marine and Coastal Fisheries: Dynamics, Management, and Ecosystem Science*, 12, 191–204. <https://doi.org/10.1002/mcf2.10117>
- National Oceanic and Atmospheric Administration. (2025). *Tide predictions for Cedar Key, FL (station 8727520)*. NOAA Tides & Currents. <https://tidesandcurrents.noaa.gov/stationhome.html?id=8727520>
- Nieto Ferreira, R., Nissenbaum, M. R., & Rickenbach, T. M. (2018). Climate change effects on summertime precipitation organization in the southeast United States. *Atmospheric Research*, 214, 348–363. <https://doi.org/10.1016/j.atmosres.2018.08.012>
- Oyster Metrics Workgroup. (2011). *Restoration goals, quantitative metrics and assessment protocols for evaluating success on restored oyster reef sanctuaries*. Sustainable Fisheries Goal Implementation Team of the Chesapeake Bay Program.
- Petes, L. E., Brown, A. J., & Knight, C. R. (2012). Impacts of upstream drought and water withdrawals on the health and survival of downstream estuarine oyster populations. *Ecology and Evolution*, 2, 1712–1724. <https://doi.org/10.1002/ece3.291>
- Pine, W. E., III, Brucker, J., Davis, M., Geiger, S., Gandy, R., Shantz, A., Stewart Merrill, T., & Camp, E. V. (2023). Collapsed oyster populations in large Florida estuaries appear resistant to restoration using traditional cultching methods—Insights from ongoing efforts in multiple systems. *Marine and Coastal Fisheries: Dynamics, Management, and Ecosystem Science*, 15, Article e10249. <https://doi.org/10.1002/mcf2.10249>
- Pine, W. E., III, Johnson, F. A., Frederick, P. C., & Coggins, L. G. (2022). Adaptive management in practice and the problem of application at multiple scales—Insights from oyster reef restoration on Florida’s Gulf coast. *Marine and Coastal Fisheries: Dynamics, Management, and Ecosystem Science*, 14, Article e10192. <https://doi.org/10.1002/mcf2.10192>
- Pine, W. E., III, Sinnickson, D., Coleman, T. S., & Frederick, P. (2026). Assessing oyster size distributions within intertidal eastern oyster *Crassostrea virginica* (Gmelin, 1791) populations across restoration sites, harvest zones, and spatial locations in the Big Bend of Florida. *Journal of Shellfish Research*, 44, 479–494. <https://doi.org/10.2983/035.044.0313>
- Pine, W. E., III, Walters, C. J., Camp, E. V., Bouchillon, R., Ahrens, R., Sturmer, L., & Berrigan, M. E. (2015). The curious case of Eastern Oyster *Crassostrea virginica* stock status in Apalachicola Bay, Florida. *Ecology and Society*, 20, Article 46. <https://doi.org/10.5751/ES-07827-200346>
- Powers, S. P., & Grabowski, J. H. (2023). Changes in water flow alter community dynamics in oyster reefs. *Ecosphere*, 14, Article e4405. <https://doi.org/10.1002/ecs2.4405>
- Powers, S. P., Peterson, C. H., Grabowski, J. H., & Lenihan, H. S. (2009). Success of constructed oyster reefs in no-harvest sanctuaries: Implications for restoration. *Marine Ecology Progress Series*, 389, 159–170. <https://doi.org/10.3354/meps08164>
- Raabe, E. A., Streck, A. E., & Stumpf, R. P. (2004). *Historic topographic sheets to satellite imagery—A methodology for evaluating coastal change in Florida’s Big Bend tidal marsh* (Open-File Report 2002-211). U.S. Geological Survey.
- Raabe, E. A., & Stumpf, R. P. (2016). Expansion of tidal marsh in response to sea-level rise: Gulf coast of Florida, USA. *Estuaries and Coasts*, 39, 145–157. <https://doi.org/10.1007/s12237-015-9974-y>
- Radabaugh, K. R., Geiger, S. P., & Moyer, R. P. (Eds.). (2019). *Oyster integrated mapping and monitoring program report for the state of Florida* (Technical Report 22). Florida Fish and Wildlife Conservation Commission, Fish and Wildlife Research Institute. <https://archive.myfwc.com/archive/Research/Habitat/Coastal-Wetlands/OIMMP/oimmp-report-2019.pdf>
- R Development Core Team. (2019). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.
- Ridge, J. T., Rodriguez, A. B., & Fodrie, F. J. (2017). Evidence of exceptional oyster-reef resilience to fluctuations in sea level. *Ecology and Evolution*, 7, 10409–10420. <https://doi.org/10.1002/ece3.3473>
- Ridge, J. T., Rodriguez, A. B., Fodrie, F. J., Lindquist, N. L., Brodeur, M. C., Coleman, S. E., Grabowski, J. H., & Theuerkauf, E. J. (2015). Maximizing oyster-reef growth supports green infrastructure with accelerating sea-level rise. *Scientific Reports*, 5, Article 14785. <https://doi.org/10.1038/srep14785>

- Schulte, D. M., Burke, R. P., & Lipcius, R. N. (2009). Unprecedented restoration of a native oyster metapopulation. *Science*, 325, 1124–1128. <https://doi.org/10.1126/science.1176516>
- Seavey, J. R., Pine, W. E., III, Frederick, P., Sturmer, L., & Berrigan, M. (2011). Decadal changes in oyster reefs in the Big Bend of Florida's Gulf coast. *Ecosphere*, 2, Article 114. <https://doi.org/10.1890/ES11-00205.1>
- Sinnickson, D., Chagaris, D., & Allen, M. (2021). Exploring impacts of river discharge on forage fish and predators using Ecopath with Ecosim. *Frontiers in Marine Science*, 8, Article 689950. <https://doi.org/10.3389/fmars.2021.689950>
- Smith, R. S., Cheng, S. L., & Castorani, M. C. N. (2023). Meta-analysis of ecosystem services associated with oyster restoration. *Conservation Biology*, 37, Article e13966. <https://doi.org/10.1111/cobi.13966>
- Smith, R. S., Hogan, S., Tedford, K. N., Lusk, B., Reidenbach, M. A., & Castorani, M. C. (2022). Long-term data reveal greater intertidal oyster biomass in predicted suitable habitat. *Marine Ecology Progress Series*, 683, 221–226. <https://doi.org/10.3354/meps13949>
- Smith, R. S., Lusk, B., & Castorani, M. C. N. (2022). Restored oyster reefs match multiple functions of natural reefs within a decade. *Conservation Letters*, 15, Article e12883. <https://doi.org/10.1111/conl.12883>
- Solinger, L. K., Ashton-Alcox, K. A., Powell, E. N., Hemeon, K. M., Pace, S. M., Soniat, T. M., & Poussard, L. M. (2022). Oysters beget shell and vice versa: Generating management goals for live oysters and the associated reef to promote maximum sustainable yield of *Crassostrea virginica*. *Canadian Journal of Fisheries and Aquatic Sciences*, 79, 1241–1254. <https://doi.org/10.1139/cjfas-2021-0277>
- Southwick Associates. (2021). *Opportunities for growth: A summary economic analysis of Florida's Big Bend region*. <https://www.southwickassociates.com/a-summary-economic-analysis-of-floridas-big-bend-region/>
- Sun, G. (2013). Impacts of climate change and variability on water resources in the southeast USA. In K. T. Ingram, K. Dow, L. Carter, & J. Anderson (Eds.), *Climate of the south-east United States: Variability, change, impacts, and vulnerability* (pp. 210–236). Island Press. https://www.srs.fs.usda.gov/pubs/ja/2013/ja_2013_sun_002.pdf
- Thieurmél, B., & Elmarhraoui, A. (2022). *Suncalc: Compute sun position, sunlight phases, moon position and lunar phase* (R package version 0.5.1) [Computer software]. <https://CRAN.R-project.org/package=suncalc>
- Thorley, J., Miller, L., & Fleishman, A. (2021). *Rtide: Tide heights* (R package version 0.0.9) [Computer software]. <https://CRAN.R-project.org/package=rtide>
- U.S. Geological Survey. (2025). *Suwannee River near Wilcox, Florida (USGS 02323500)*. U.S. Geological Survey, National Water Information System. Retrieved January 4, 2026, from <https://waterdata.usgs.gov/monitoring-location/02323500/>
- Vehtari, A., Gabry, J., Magnusson, M., Yao, Y., Bürkner, P., Paananen, T., & Gelman, A. (2023). *loo: Efficient leave-one-out cross-validation and WAIC for Bayesian models* (R package version 2.6.0) [Computer software]. <https://mc-stan.org/loo/>
- Walker, B. H., Carpenter, S. R., Rockstrom, J., Crépin, A.-S., & Peterson, G. D. (2012). Drivers, “slow” variables, “fast” variables, shocks, and resilience. *Ecology and Society*, 17, Article 30. <https://doi.org/10.5751/ES-05063-170330>
- Walters, C. J. (1986). *Adaptive management of renewable resources*. Macmillan.
- Walters, C. J., & Holling, C. S. (1990). Large-scale management experiments and learning by doing. *Ecology*, 71, 2060–2068. <https://doi.org/10.2307/1938620>
- Walters, C. J., & Maguire, J.-J. (1996). Lessons for stock assessment from the Northern Cod collapse. *Reviews in Fish Biology and Fisheries*, 6, 125–137. <https://doi.org/10.1007/BF00182340>
- Wright, E. E., Hine, A. C., Goodbred, S. L., & Locker, S. D. (2005). The effect of sea-level and climate change on the development of a mixed siliciclastic carbonate, deltaic coastline: Suwannee River, Florida, USA. *Journal of Sedimentary Research*, 75, 621–635. <https://doi.org/10.2110/jsr.2005.051>
- zu Ermgassen, P., Hancock, B., DeAngelis, B., Greene, J., Schuster, E., Spalding, M., & Brumbaugh, R. (2016). *Setting objectives for oyster habitat restoration using ecosystem services: A manager's guide*. The Nature Conservancy. <https://oceanwealth.org/wp-content/uploads/2016/06/Setting-Objectives-for-Oyster-Habitat-Restoration-Using-Ecosystem-Service.pdf>

Appendix A: Observer training

To minimize error and count bias, all observers received advance training on field procedures using a simulated oyster reef. The simulator used paper bags to represent substrate (rock/cultch), mock paper oysters glued to the bags, and marked bags to mimic other reef organisms. Observers conducted replicate passes (≥ 3 passes) on the artificial reef; results were then compared to illustrate the range of counts generated by experienced versus inexperienced observers. During training, an experienced crew member reviewed every oyster counted on a transect to confirm key rules, including counting only oysters within the belt defined by strings between the H frames. At the end of each session, all paper oysters were removed and tallied to establish the true count, and each observer's pass was compared to this benchmark.

Appendix B: Model structure overview

We modeled counts of live oysters by using a negative-binomial distribution with a log link, which is appropriate for overdispersed count data. An offset for transect length was included to account for variation in sampling effort across sites.

The general model form is

$$Y_{ijk} \sim \text{NegBinomial}(\mu_{ijk}, \phi),$$

$$\log(\mu_{ijk}) = \eta_{ijk} + \log(\text{Length}_{ijk}),$$

where Y_{ijk} is the count of live oysters at transect k , site j , and time i ; μ_{ijk} is the expected count; ϕ is the dispersion parameter; η_{ijk} is the linear predictor; Length_{ijk} is the transect length (m), and $\log$ denotes the natural logarithm.

Question 1: Temporal trends across reef zones

This model assessed whether oyster counts changed differently over time in inshore, nearshore, and offshore reef zones:

$$\log(\mu_{ijk}) = \alpha + \beta_1 \cdot \text{Zone}_j + \beta_2 \cdot \text{Period}_i + \beta_3 \cdot (\text{Zone}_j \cdot \text{Period}_i) + \log(\text{Length}_{ijk}).$$

No random effects were included.

Question 2: Repeated-measures postrestoration

This model evaluated whether oyster counts increased postrestoration compared to prerestoration baselines within the same sites, accounting for repeated measures:

$$\log(\mu_{ijk}) = \alpha + \beta_1 \cdot \text{Strata}_j + \beta_2 \cdot \text{Period}_i + \beta_3 \cdot (\text{Strata}_j \cdot \text{Period}_i) + u_j + \log(\text{Length}_{ijk}),$$

$$u_j \sim N(0, \sigma_{\text{site}}^2),$$

where Strata indicates restored or unrestored status and u_j is a random intercept for site.

Question 3: Lone Cabbage restored versus unrestored · harvest status

This model focused on the Lone Cabbage site only, testing whether harvest openings differentially affected restored versus unrestored reef:

$$\log(\mu_{ijk}) = \alpha + \beta_1 \cdot \text{Harvest}_j + \beta_2 \cdot \text{Strata}_j + \beta_3 \cdot (\text{Harvest}_j \cdot \text{Strata}_j) + u_i + \log(\text{Length}_{ijk}),$$

$$u_i \sim N(0, \sigma_{\text{period}}^2),$$

where Harvest is a binary indicator (open or closed) and u_i is a random intercept for period.

Question 4: Covariate models

We fitted two sets of models to address covariates of management interest and their potential effects on oyster counts.

Question 4a: Covariate · site interaction model

We evaluated how standardized covariates (river discharge, landings, and trips) interacted with restoration status and harvest status to explain oyster density while controlling for period-level variation:

$$\log(\mu_{ijk}) = B_{1[i]} + B_{\text{covariate}} \cdot X_{ijk} + \beta_{\text{interaction}} \cdot (\text{SiteRock}_j \cdot X_{ijk}) + \log(\text{Length}_{ijk}) + \gamma_{\text{period}[i]},$$

where X_{ijk} is the standardized covariate of interest (discharge, landings, or trips), $B_{1[i]}$ is a categorical variable for site $\times$ restoration type (no intercept), $\beta_{\text{interaction}}$ captures the covariate $\times$ restoration status interaction, and $\gamma_{\text{period}[i]} \sim N(0, \sigma_{\text{period}}^2)$ and is a random effect for time.

Question 4b: Simplified rock · harvest model (rock_fish)

To directly compare restoration and harvest status combinations, we fitted a simplified Bayesian negative-binomial model that estimates separate means for each combination of restoration (yes or no) and harvest status (open or closed):

$$\log(\mu_{ijk}) = B_{\text{rockfish}[i]} + \log(\text{Length}_{ijk}) + \gamma_{\text{period}[i]},$$

where $B_{\text{rockfish}[i]}$ is a categorical predictor indexing all combinations of restoration and harvest status (restored and open, restored and closed, unrestored and open, unrestored and closed). Each level is estimated independently by specifying a no-intercept model structure (i.e., $0 + B_{\text{rockfish}[i]}$). The term

Table B1 Period, years, months in which oyster population sampling took place, and major project activity as part of the Lone Cabbage Reef restoration and monitoring program.

Period	Years	Months during which sampling was conducted in period	Project activity
1	2010	May, Jul, Aug	Prerestoration
2	2010	Oct, Dec	Prerestoration
3	2011	Jul	Prerestoration
6	2012	Oct, Nov	Prerestoration
7	2013	Apr	Prerestoration
7	2013	Sep, construction	Lone Cabbage Reef pilot project
10	2014–2015	Oct, Nov, Dec, Jan	Lone Cabbage Reef pilot project
11	2015	May, Jun	Lone Cabbage Reef pilot project
16	2017–2018	Nov, Jan	Lone Cabbage Reef pilot project
17	2018	Jul–Oct, construction and structure assessment	Lone Cabbage Reef restoration construction
18	2018–2019	Nov, Dec, Jan, Feb, Mar	Lone Cabbage Reef restoration
19	2019	May, Jun, Jul, Aug	Lone Cabbage Reef restoration
20	2019–2020	Oct, Nov, Dec, Jan	Lone Cabbage Reef restoration
22	2020–2021	Nov, Dec, Jan, Feb	Lone Cabbage Reef restoration
24	2021–2022	Nov, Dec, Jan, Feb	Lone Cabbage Reef restoration
26	2022–2023	Oct, Nov, Dec, Jan, Feb	Lone Cabbage Reef restoration

Table B2 Summary of transect counts by semiannual period, site, restoration status, and harvest status at Lone Cabbage. Columns are as follows: Period = semiannual monitoring index; Site = inshore (I), nearshore (N), or offshore (O); Restoration = unrestored (N) or restored (Y); Harvest = closed to harvest (N) or open to harvest (Y); Number of transects = count of transects sampled in the cell; and Sum transect length = total surveyed transect length (m). Period is derived from the sample date: Period 1 is April–September 2010; period 2 is October 2010–March 2011; and the index then increases by 1 every half-year (April–September, then October–March), with January–March assigned to the prior October–March period.

Period	Site	Restoration	Harvest	Number of transects	Sum transect length (m)
1	I	N	N	5	91.9
1	I	N	Y	8	198.1
1	N	N	Y	16	384.8
1	O	N	N	3	57.2
1	O	N	Y	10	354.0
2	I	N	N	5	91.9
2	I	N	Y	4	98.8
2	N	N	Y	12	258.0
2	O	N	N	3	55.9
2	O	N	Y	6	248.5
3	I	N	N	5	91.1
3	I	N	Y	3	64.3
3	N	N	Y	8	172.7
3	O	N	N	3	56.0
3	O	N	Y	6	235.0

(Continued)

Table B2 (Continued)

Period	Site	Restoration	Harvest	Number of transects	Sum transect length (m)
6	I	N	N	5	103.5
6	I	N	Y	6	152.0
6	N	N	Y	13	315.8
6	O	N	N	3	75.0
6	O	N	Y	6	272.5
7	O	N	N	8	528.0
10	O	N	N	4	255.7
10	O	Y	N	4	255.9
11	O	N	N	4	254.9
11	O	Y	N	4	256.2
16	O	N	N	4	264.0
16	O	Y	N	4	264.0
18	I	N	N	18	570.9
18	I	N	Y	17	502.7
18	N	N	Y	9	225.6
18	O	Y	N	13	976.6
18	O	Y	Y	4	383.7
19	I	N	N	5	92.6
19	I	N	Y	8	226.0
19	N	N	Y	13	330.6
19	O	N	Y	9	294.9
20	I	N	N	18	594.8
20	I	N	Y	12	396.7
20	N	N	Y	5	205.2
20	O	Y	N	7	926.2
20	O	Y	Y	5	463.6
22	I	N	N	20	546.5
22	I	N	Y	10	303.4
22	N	N	Y	5	222.8
22	O	Y	N	9	1,323.7
22	O	Y	Y	5	1,138.3
24	I	N	N	19	521.2
24	I	N	Y	13	367.9
24	N	N	Y	2	44.0
24	O	Y	N	11	1,425.3
24	O	Y	Y	3	700.2
26	I	N	N	6	216.1
26	I	N	Y	12	269.9
26	N	N	Y	3	72.4
26	O	Y	N	5	728.7
26	O	Y	Y	5	871.4

Table B3 Period-specific posterior predictions for the postrestoration repeated-measures analysis (Question 2): mean (μ) live oysters (≥ 18 mm shell height) per square meter by monitoring period, with lower and upper limits of the 95% credible interval (CrI). Predictions include the period random effect and use an offset for log(transect length).

Period	Expected μ per m ²	95% CrI for μ per m ²	
		Lower	Upper
18	91	43	185
20	132	72	232
22	156	89	272
24	199	115	356
26	244	134	433

$\log(\text{Length}_{ijk})$ is an offset term that accounts for variation in transect lengths, ensuring that model predictions represent oyster density (oysters per meter of transect) rather than raw counts; $\gamma_{\text{period}[i]} \sim N(0, \sigma_{\text{period}}^2)$ and is a random intercept for each period that accounts for time-based variability in oyster counts.

This model facilitates posterior prediction and contrasts between combinations of restoration and harvest status that are relevant to management.

Predicted values were back-transformed from the log scale using the exponential function and then were converted to oyster density per square meter by dividing by the transect width (0.1524 m).

All models were fitted using four chains of 3,000 iterations each, with 1,500 warmup iterations. Convergence was assessed using convergence diagnostics (R-hat; required ≤ 1.01) and effective sample sizes; all models met these criteria.

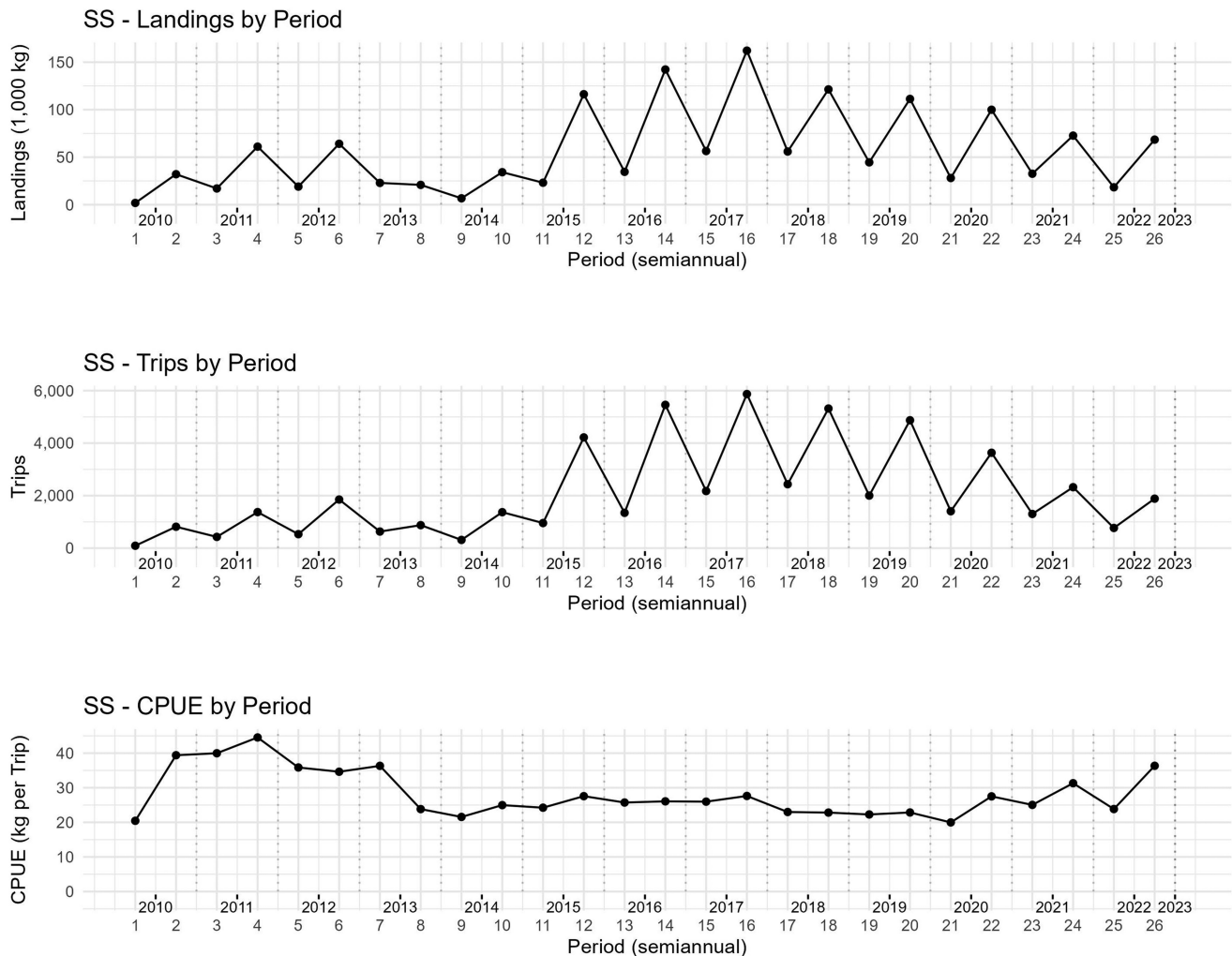


Figure B1 Commercial oyster fishery landings (meat pounds, by convention), trips, and CPUE for Suwannee Sound (SS; Taylor, Dixie, and Levy counties combined) by period. The sawtooth appearance is a result of seasonal fishery closures.

Table B4 Offshore restoration and harvest summary for live-oyster density, contrasts, and ratios. Values are population-average posterior summaries for live-oyster density of oysters ≥ 18 mm on offshore reefs, expressed as oysters/m² using a transect width of 0.1524 m. The table includes group estimates, differences, and ratios used to summarize questions 3 and 4. Group codes are as follows: O_N = offshore unrestored, O_Y = offshore restored, O_Y_N = offshore restored and closed to harvest, and O_Y_Y = offshore restored and open to harvest. Comparisons include restoration contrasts between offshore restored and unrestored reefs, O_Y - O_N and O_Y ÷ O_N, and harvest-status contrasts on restored offshore reefs, Closed - open and Open ÷ closed. Comparison identifies the management comparison being summarized; Quantity identifies whether the row is a group-level estimate or a contrast; Group identifies the paired comparison role, where A and B are the two groups being compared, A - B is the additive difference, and Ratio is the proportional comparison; Label provides the display label for the group or contrast; Periods_A and Periods_B identify the semiannual periods informing each side of the comparison; Overlap identifies periods shared by both sides of the comparison; Statistic identifies the posterior summary statistic; and Value gives the corresponding estimate. Group-level density estimates and additive differences are reported on the oysters/m² scale, with positive differences indicating higher density for A relative to B. Ratios are unitless, with values greater than 1 indicating higher density for the numerator group. Estimates are from negative-binomial generalized linear mixed models with a log link, offset of log(transect length), and a period × restoration random intercept; random effects are excluded. Periods are semiannual indices, where period 1 = April–September 2010, period 2 = October 2010–March 2011, and each subsequent period advances by one half-year. For difference summaries, credible intervals overlapping 0 indicate uncertainty in the direction of the contrast; for ratio summaries, credible intervals overlapping 1 indicate uncertainty in the direction of the proportional difference.

Comparison	Quantity	Group	Label	Periods_A	Periods_B	Overlap	Statistic	Value
Harvest status on offshore restored	Level	A	Closed (O_Y_N)	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Mean	88
Harvest status on offshore restored	Level	A	Closed (O_Y_N)	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Lower CrI	25
Harvest status on offshore restored	Level	A	Closed (O_Y_N)	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Upper CrI	166
Harvest status on offshore restored	Level	B	Open (O_Y_Y)	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Mean	59
Harvest status on offshore restored	Level	B	Open (O_Y_Y)	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Lower CrI	16
Harvest status on offshore restored	Level	B	Open (O_Y_Y)	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Upper CrI	121
Harvest status on offshore restored	Contrast	A - B	Closed - open	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Mean	29
Harvest status on offshore restored	Contrast	A - B	Closed - open	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Lower CrI	-5
Harvest status on offshore restored	Contrast	A - B	Closed - open	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Upper CrI	78
Harvest status on offshore restored	Contrast	Ratio	Open ÷ closed	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Mean	0.69
Harvest status on offshore restored	Contrast	Ratio	Open ÷ closed	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Lower CrI	0.4

(Continued)

Table B4 (Continued).

Comparison	Quantity	Group	Label	Periods_A	Periods_B	Overlap	Statistic	Value
Harvest status on offshore restored	Contrast	Ratio	Open ÷ closed	10, 11, 16, 18, 20, 22, 24, 26	18, 20, 22, 24, 26	18, 20, 22, 24, 26	Upper CrI	1.07
Restoration effect at offshore reefs	Level	A	Restored (O_Y)	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Mean	105
Restoration effect at offshore reefs	Level	A	Restored (O_Y)	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Lower CrI	45
Restoration effect at offshore reefs	Level	A	Restored (O_Y)	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Upper CrI	194.2
Restoration effect at offshore reefs	Level	B	Unrestored (O_N)	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Mean	21
Restoration effect at offshore reefs	Level	B	Unrestored (O_N)	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Lower CrI	6.4
Restoration effect at offshore reefs	Level	B	Unrestored (O_N)	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Upper CrI	45.3
Restoration effect at offshore reefs	Contrast	A – B	Restored – unrestored	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Mean	84
Restoration effect at offshore reefs	Contrast	A – B	Restored – unrestored	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Lower CrI	21.9
Restoration effect at offshore reefs	Contrast	A – B	Restored – unrestored	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Upper CrI	174.7
Restoration effect at offshore reefs	Contrast	Ratio	Offshore restored ÷ offshore unrestored	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Mean	6
Restoration effect at offshore reefs	Contrast	Ratio	Offshore restored ÷ offshore unrestored	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Lower CrI	1.68
Restoration effect at offshore reefs	Contrast	Ratio	Offshore restored ÷ offshore unrestored	10, 11, 16, 18, 20, 22, 24, 26	1, 2, 3, 6, 7, 10, 11, 16	10, 11, 16	Upper CrI	17.35

Table B5 The difference ($\pm$ SE) in expected log predictive density (Δ ELPD) relative to the top model (more negative = worse) for each model fit in research question 4. The $n(\text{Pareto } k > 0.7)$ column shows the counts of observations with high Pareto k -values (potential importance sampling issues; fewer is better). Models are negative-binomial generalized linear mixed models (log link) with an offset of log(transect length) and a period $\times$ restoration random intercept, all fitted to the same observations. Abbreviations are as follows:

AD = mean daily discharge, TD = total discharge, Trips = commercial fishing trips, Landings = commercial landings; Lag1 = 1-year lag (two semiannual periods), and Lag2 = 2-year lag (four semiannual periods). Base is the site-only model (no covariate), and Fish is the site $\times$ restoration $\times$ harvest categorical model, which was used to compare restored offshore reefs that were closed versus open to harvest.

Model	Δ ELPD from best ($\pm$ SE)	$n(\text{Pareto } k > 0.7)$
TDLag2	-16.50 ± 6.94	4
ADLag2	-16.43 ± 6.96	2
Fish	-14.69 ± 6.62	1
Site-only model (base)	-14.66 ± 6.59	2
TD	-12.22 ± 6.08	3
AD	-11.74 ± 6.08	1
TDLag1	-11.09 ± 7.06	3
ADLag1	-10.86 ± 7.07	2
Trips	-6.32 ± 5.15	3
Landings	-4.73 ± 3.29	2
TripsLag1	-4.55 ± 4.56	2
LandingsLag1	-3.21 ± 4.19	2
TripsLag2	-1.73 ± 1.25	2
LandingsLag2	0.00 ± 0.00	3

Appendix C: Key elements of adaptive management of the Lone Cabbage Reef restoration

The Lone Cabbage Reef restoration incorporated adaptive management—a structured approach in which management actions are used to test hypotheses, evaluate outcomes, and update decisions (Walters, 1986; Walters & Holling, 1990). This framework influenced project design, implementation, and monitoring under uncertainty (Pine et al., 2022). Rather than applying a fixed restoration design, the project included a pilot phase, targeted experiments, and postconstruction monitoring to evaluate performance and refine subsequent actions. In contrast, several restoration efforts in Florida and elsewhere have relied on prescriptive designs that limited opportunities to evaluate causes of decline or determinants of restoration success (Bersoza Hernández et al., 2018; Pine

et al., 2023). Key adaptive management elements of the Lone Cabbage Reef project are summarized below.

Dual-control challenge

Adaptive management requires balancing implementation with information generation (Walters, 1986). The pilot phase at Lone Cabbage Reef tested location, substrate, and elevation to identify combinations that supported oyster recruitment (Frederick et al., 2016). Following encouraging results, the full restoration adopted a standardized design to accelerate construction (Aufmuth et al., 2025; Pine et al., 2022). This increased implementation efficiency but reduced opportunities for continued experimentation. Given the urgency of reef decline (Moore et al., 2020; Seavey et al., 2011) and the cost of mobilization (Pine et al., 2022), this represented a pragmatic trade-off between learning and rapid restoration.

Deliberate experimentation

The pilot phase addressed a central uncertainty: whether recovery was limited primarily by substrate availability or by larval supply. Results supported substrate limitation under suitable environmental conditions and adequate larval input (Frederick et al., 2016), providing justification for scaling up restoration. If larval supply or environmental constraints had dominated, a different strategy would have been required.

Site selection

Selection of Lone Cabbage Reef was informed by harvester knowledge, reconstructed historical maps (Raabe et al., 2004), and aerial photography documenting gradual degradation rather than abrupt loss (Seavey et al., 2011). The reef's proximity to the Suwannee River may buffer salinity extremes, and nearby healthy reefs and subtidal harvest areas likely provide local larval sources (Grizzle et al., 2021). Site selection was evaluated during the pilot phase to confirm suitability (Frederick et al., 2016).

Reef design

Restoration elevations were designed to match nearby robust intertidal reefs, with final target elevations extending to approximately -0.36 m NAVD88 (North American Vertical Datum of 1988; Aufmuth et al., 2025). Higher-elevation reefs in other systems support greater oyster densities through effects on hydrodynamics, larval settlement, and predation (Colden et al., 2017; Grabowski, 2004; Lenihan, 1999; Powers & Grabowski, 2023; Schulte et al., 2009; Smith, Hogan, et al., 2022; Smith, Lusk, & Castorani, 2022). These considerations informed design targets.

Substrate selection

The pilot phase compared candidate cultch materials and identified dolomite limestone—a naturally occurring substrate common in nearby Waccasassa Bay—as suitable for oyster settlement and persistence (Frederick et al., 2016). Because inappropriate substrate choice is a common cause of restoration failure (Goelz et al., 2020; Pine et al., 2023), this evaluation reduced implementation risk.

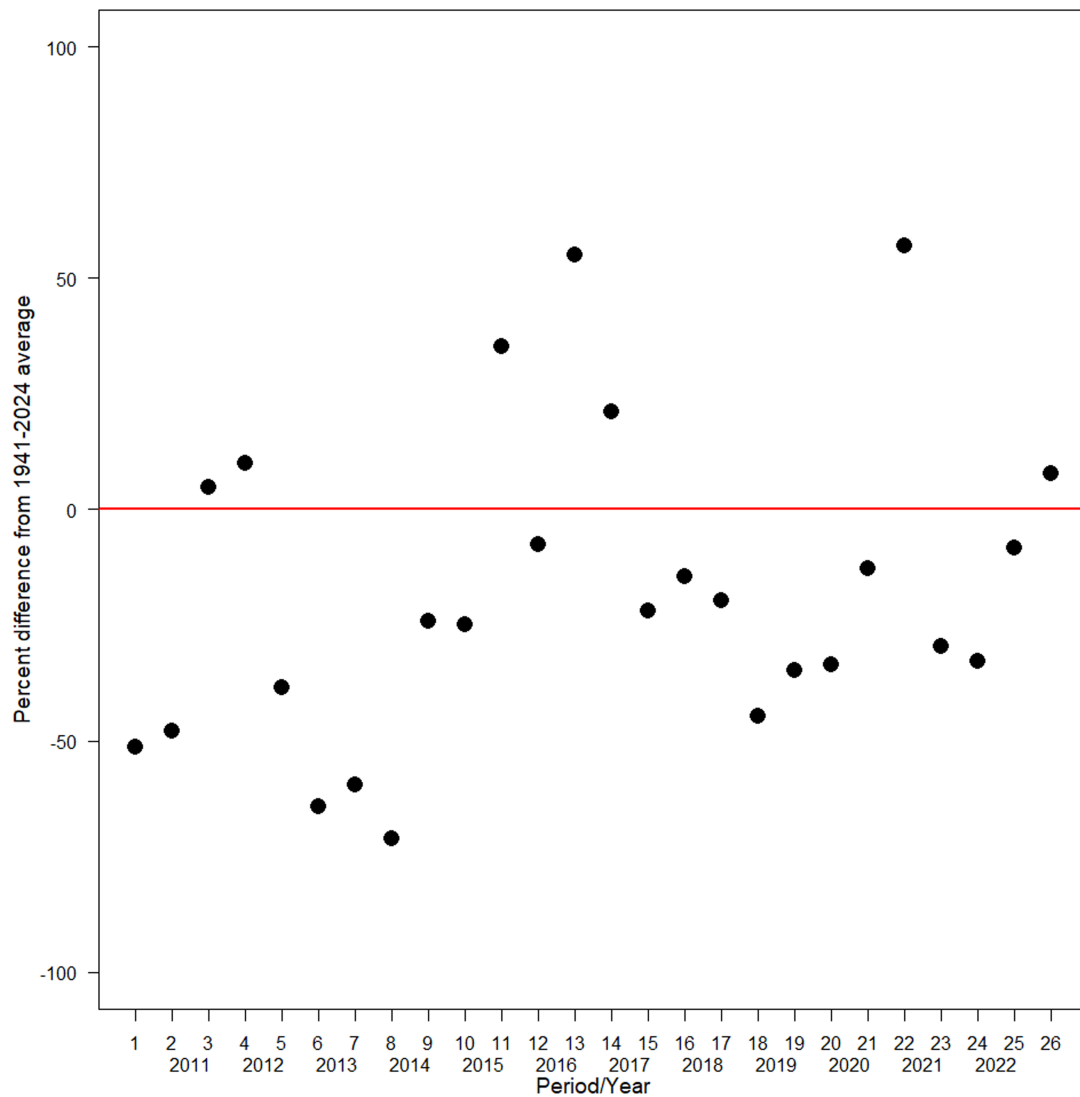


Figure B2 Suwannee River discharge: percent difference between the mean daily average for each period (black dots) and the mean daily average for the entire period of record (1941–March 2024; red line). The mean daily average discharge for the period of record is 9,671.7 ft³ (272.3 m³) by convention. Discharge was measured at U.S. Geological Survey gauge 02323500 (near Wilcox, Florida; [U.S. Geological Survey, 2025](#)).

Monitoring and measurable success

Monitoring was incorporated to evaluate recruitment and population response. Larval settlement occurred within weeks of construction, and multiple cohorts were observed over 5 years. [Pine et al. \(2026\)](#) documented shifts in size structure toward nearby healthy reefs within 2 years. Together with the density increases reported here, these results indicate

successful recruitment and population growth consistent with oyster restoration performance metrics ([Oyster Metrics Workgroup, 2011](#); [Pine et al., 2023](#)).

Taken together, these design and monitoring elements reduced key uncertainties, and they provide a practical framework for planning and evaluating oyster restoration in substrate-limited systems.

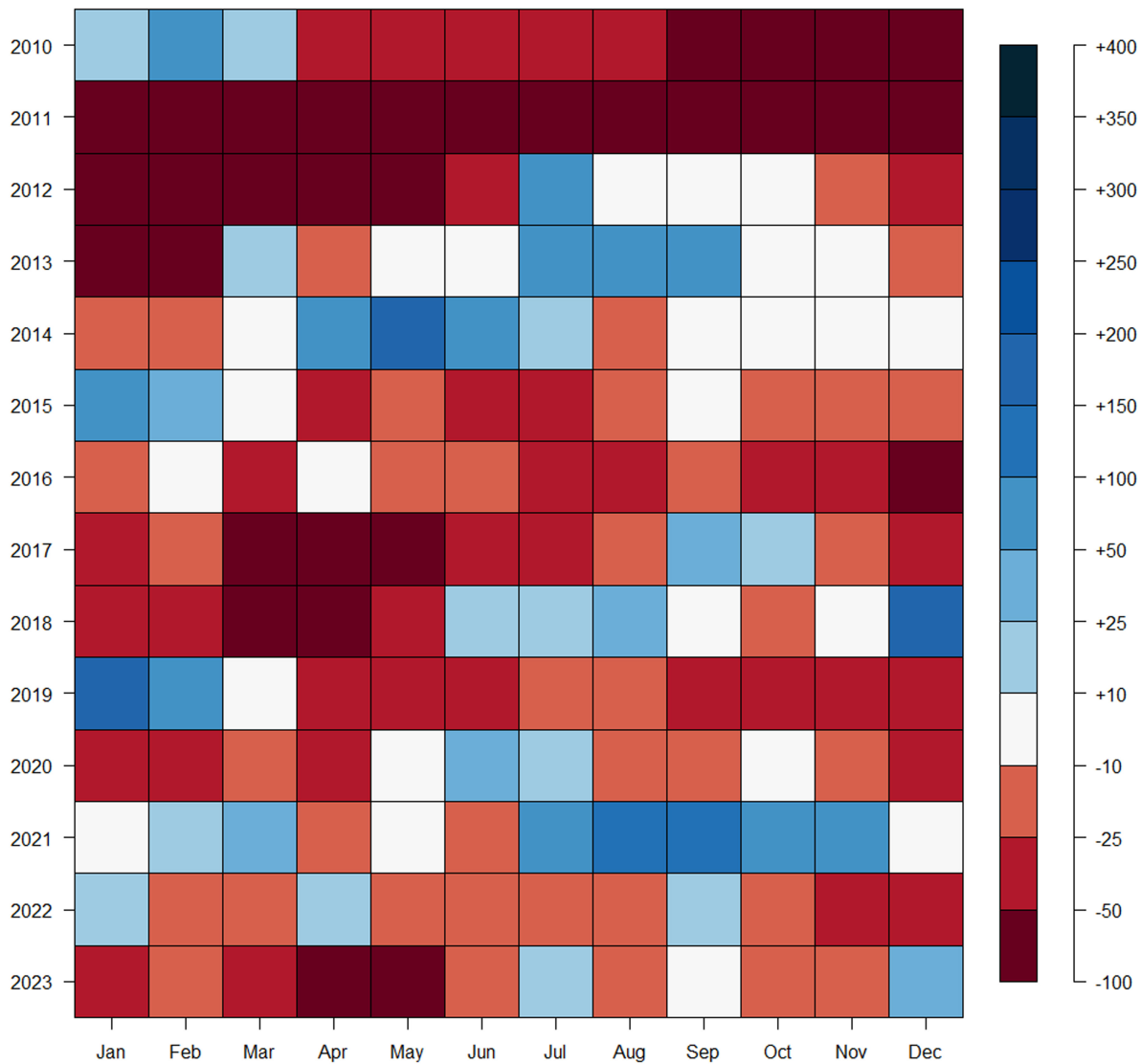


Figure B3 Suwannee River discharge: percent difference between the monthly average for a given year (2010–2023) and the average for that month for the period of record (1941–March 2024). Measurements were taken at U.S. Geological Survey gauge 02323500 (near Wilcox, Florida; [U.S. Geological Survey, 2025](#)).

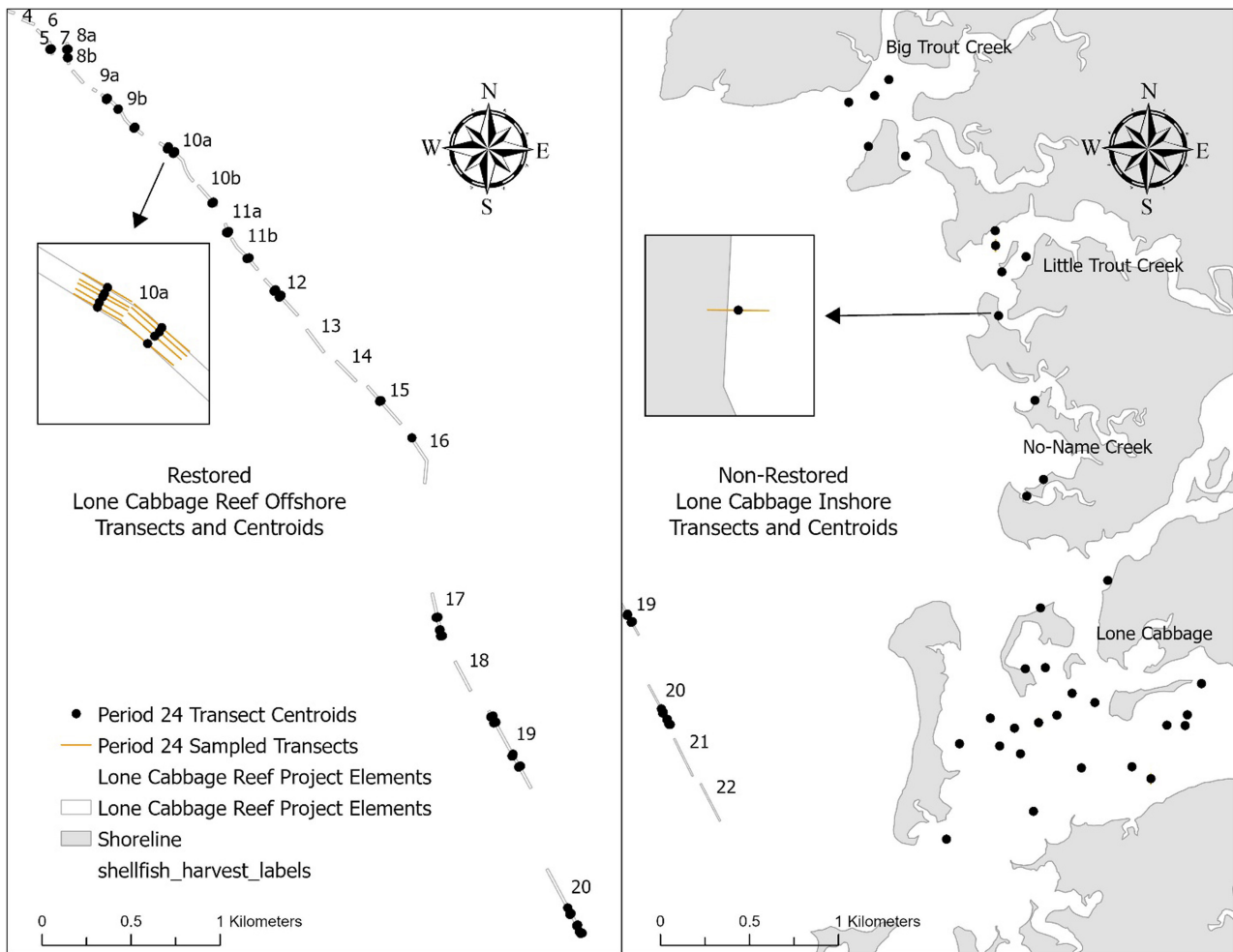


Figure B4 Detail of restored Lone Cabbage Reef elements, with the centroids of the sample locations (left panel, dots). The inset within the left panel demonstrates the transects that were used for oyster counts within reef element 10a as a representative example. The right panel depicts inshore and nearshore locations where oyster counts took place during period 24. The inset map within the right panel demonstrates an example transect.



Figure B5 Monitoring at a restored site on Lone Cabbage Reef in January 2023 by participants from the University of Florida, the Florida Fish and Wildlife Conservation Commission, and commercial fishing industry partners. Oyster clusters are visible on and between placed rocks, and gaps without rock are evident (top). The bottom photo highlights the contrast between areas without rock (left side of image) and with rock (right side of image, where observers are looking). (Photos courtesy of C. Palmer.)